Polynomial mixing for the white-forced Navier–Stokes system in the whole space

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Abstract

We study the mixing properties of the two-dimensional Navier–Stokes system on the whole space $\mathbb{R}^2$ perturbed by an additive white-in-time noise. Under a suitable non-degeneracy assumption on the noise, we establish the uniqueness of the stationary measure and prove polynomial mixing in the dual-Lipschitz metric. This provides the first ergodicity and mixing result for the white-forced Navier–Stokes system in an unbounded domain.

The main difficulty arises from the lack of compactness and spectral properties that are essential in bounded settings. To overcome this, we use a space-time weight function of the Muckenhoupt A_2 class, which allows us to control the behavior of solutions at infinity and to derive a suitable Foiaş–Prodi type stability estimate. In addition, the proof combines the coupling method, weighted growth bounds for trajectories, and an estimate for the Leray projector acting on weighted spaces.

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Keywords: Stochastic Navier–Stokes system, polynomial mixing, coupling method, Foiaş–Prodi estimate, weighted growth estimates

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1 Introduction

Over the past two decades, substantial progress has been made in studying the ergodic properties of dissipative PDEs driven by white-in-time noise. Most existing results concern PDEs on bounded domains, where one can take advantage of compactness properties such as compact Sobolev embeddings and spectral properties, including the discreteness of the Laplacian spectrum. Among these equations, the Navier–Stokes (NS) system has been studied most extensively because of its fundamental role in fluid dynamics and turbulence. The first key results were obtained in [FM95, EMS01, KS02, Hai02, HM06, Oda06], while further developments are presented in the book [KS12] and the surveys [KS17, Deb13].

In contrast, the situation in unbounded domains is considerably more complicated because the compactness and spectral properties mentioned above no longer hold. As a result, even the existence of a stationary measure is non-trivial. In unbounded Poincaré-type domains and for the damped NS system in the whole space $\mathbb{R}^2$, the existence of a stationary measure was established only relatively recently [BL06], while questions of uniqueness and mixing remained open (see Open Problems 1 and 2 in [KS17]). In this work, we present the first results on ergodicity and mixing for the NS system in unbounded domains under additive white-in-time noise. Specifically, we study the dynamics of the following damped randomly forced NS system:

$$\begin{cases} \partial_t u + (u\cdot\nabla)u - \nu\Delta u + au + \nabla p = h + \eta, & x \in \mathbb{R}^2, \\ \operatorname{div} u = 0, \\ u|_{t=0} = u_0, \end{cases} \tag{1.1}$$

where $a, \nu > 0$ are respectively the damping parameter and the kinematic viscosity, u is the velocity field, and p is the pressure. We consider the system (1.1) in the space of divergence-free vector fields

$$H := \{u \in L^2(\mathbb{R}^2; \mathbb{R}^2) \mid \operatorname{div} u = 0\} \tag{1.2}$$

endowed with the inner product $\langle u, v\rangle$ and the norm $\|u\|^2 := \langle u, u\rangle$ inherited from $L^2 := L^2(\mathbb{R}^2; \mathbb{R}^2)$. The external force consists of a deterministic component h and a stochastic component

$$\eta(t) := \partial_t \sum_{j=1}^{\infty} b_j \beta_j(t) e_j, \tag{1.3}$$

where $\{\beta_j\}_{j=1}^{\infty}$ are independent standard Brownian motions defined on a probability space $(\Omega, \mathscr{F}, \mathscr{F}_t, \mathbb{P})$ satisfying the usual conditions (e.g., see Definition 2.25 in [KS14]), $\{b_j\}_{j=1}^{\infty}$ are real numbers with

$$\mathcal{B}_0 := \sum_{j=1}^{\infty} b_j^2 < \infty, \tag{1.4}$$

and $\{e_j\}_{j=1}^{\infty}$ is an orthonormal basis in H. Under these assumptions, the equation (1.1) is globally well-posed in H and defines a Markov process.

For any $m > 0$, let $H^m := H^m(\mathbb{R}^2; \mathbb{R}^2) \cap H$ denote the usual Sobolev space of divergence-free vector fields. Our main result is stated as follows.

Main Theorem. *Assume that*

$$\varphi e_j \in H^1, \qquad e_j \in H^2, \qquad j \geq 1 \tag{1.5}$$

with $\varphi(x) := \sqrt{|x|^2 + 1}$, *and*

$$\mathcal{B}_\varphi := \sum_{j=1}^{\infty} b_j^2 \left(\|\varphi e_j\|^2 + \|\varphi \operatorname{curl} e_j\|^2\right) < \infty. \tag{1.6}$$

Then, there is an integer $N \geq 1$ *such that the stochastic NS system* (1.1) *admits a unique stationary measure* μ, *provided that*

$$b_j \neq 0 \quad \text{for } 1 \leq j \leq N \tag{1.7}$$

and h *belongs to the space spanned by the family* $\{e_1, e_2, \ldots, e_N\}$. *Moreover, for any* $q > 1$, *there is a constant* $C_q > 0$ *such that*

$$\left|\mathbb{E}(f(u(t))) - \int_H f(u)\mu(\mathrm{d}u)\right| \leq C_q \|f\|_{\mathrm{Lip}} (1+t)^{-q} \left(1 + \|u_0\|^2\right), \quad t \geq 0$$

for any initial data $u_0 \in H$ *and any bounded Lipschitz-continuous function* $f : H \to \mathbb{R}$.

It is worth noting that no restriction is imposed on the size of the parameters a and ν, and the intensity of the noise is independent of both. The assumption that h is a linear combination of $e_1, e_2, \dots, e_N$ can be relaxed to the following regularity and summability conditions:

$$h \in H^1, \qquad h\varphi \in H, \qquad \sum_{j=1}^{\infty} |\langle h, e_j \rangle| \|e_j\|_{H^1} < \infty. \tag{1.8}$$

In this case, however, the integer N in the condition (1.7) will depend on the convergence rate q toward the stationary measure μ. We refer to Remark 5.2 for further details.

In the literature, only a few works consider the problem of the uniqueness of stationary measure and mixing for randomly forced PDEs in unbounded domains. Most of these focus on Burgers-type equations perturbed by space-time homogeneous noise, using specific features of the Burgers equation, such as the Hopf–Cole transform, L^1-contraction, and the comparison principle; see [BCK14, BL19, DGR21]. The methods in these papers are specific to the Burgers-type equations and do not extend to the NS system. For the NS system with non-homogeneous bounded noise, uniqueness and exponential mixing in unbounded domains were established in [Ner22] via a controllability approach combined with the asymptotic compactness of the dynamics. When the noise is white-in-time, uniqueness and mixing results have so far been obtained only for PDEs with local nonlinearities. In [NZ24], the stochastic complex Ginzburg–Landau equation is studied, where the main idea of the proof is to quantitatively describe the spatial decay rate of solutions by introducing a space-time weight function, and use the spatial decay to compensate for the loss of compactness. In the paper [Gao24], the ideas of [NZ24] have been extended to the case of the Kuramoto–Sivashinsky equation. As for the NS system in unbounded domains driven by white-in-time noise, the existence of a stationary measure has been proven in [BL06], but until now, no results on uniqueness have been available.

The proof of the Main Theorem relies on the coupling method, as described in Section 3.1.2 in [KS12] and [Shi08], combined with weighted estimates developed in [NZ24]. Extending this approach to the case of the NS system in the whole space $\mathbb{R}^2$ poses a significant challenge, due to the interplay between the non-local nature of the equation and the asymptotic behavior of solutions at infinity. To overcome this difficulty, we introduce a space-time weight function ψ that belongs to the Muckenhoupt A_2 class with the characteristic A_2-constant $[\psi(t)]_{A_2}$ being uniformly bounded in time. As a result, an appropriate estimate for the operator $\psi\Pi$, where Π is the Leray projector, is derived, which further ensures a Foiaş–Prodi type estimate. The latter, together with an estimate for the growth of weighted parabolic energy and an application of the Girsanov theorem, enables us to verify the polynomial squeezing property, which plays a central role in the coupling argument.

The methods of this paper extend without difficulties to the case of the NS system in any Poincaré-type domain (that is, a domain bounded in one direction) equipped with Lions boundary conditions. It should be noted, however, that these

methods do not directly apply to problems with no-slip boundary conditions, as they rely on a probabilistic linear growth estimate for the vorticity – a property that fails under no-slip boundary conditions (see [KV14]).

The structure of the paper is as follows. In Section 2, we provide a detailed construction of the coupling processes and show how the proof of polynomial mixing reduces to verifying recurrence and squeezing properties. Section 3 is dedicated to deriving a growth estimate for a weighted parabolic energy functional. In Section 4, we establish a Foiaş–Prodi type estimate along with growth estimates for an auxiliary process. The recurrence and polynomial squeezing properties are shown in Section 5. Finally, the Appendix collects several technical results used throughout the paper.

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Notation

Let H be the space of divergence-free vector fields defined by (1.2), equipped with the L^2-inner product $\langle u, v\rangle$ and the corresponding norm $\|u\|^2 := \langle u, u\rangle$. For any $m > 0$, let $H^m := H^m(\mathbb{R}^2;\mathbb{R}^2)\cap H$ be the Sobolev space of divergence-free vector fields. We shall use the following notation.

$C_b(H)$−the space of bounded continuous functions $f : H \to \mathbb{R}$ with the norm

$$\|f\|_\infty := \sup_{u\in H} |f(u)|;$$

$\mathrm{Lip}(H)$−the space of bounded Lipschitz-continuous functions $f : H \to \mathbb{R}$ with

$$\|f\|_{\mathrm{Lip}} := \|f\|_\infty + \sup_{\substack{u,v\in H\\ u\neq v}} \frac{|f(u) - f(v)|}{\|u - v\|};$$

$B_H(u, R)$−the open ball in H of radius $R > 0$ centered at $u \in X$;

$\overline{B}_H(u, R)$−the closure of $B_H(u, R)$;

$\mathbb{I}_\Gamma$−the indicator function of a set $\Gamma \subset H$.

$\mathscr{B}(H)$−the Borel σ-algebra of H;

$\mathscr{P}(H)$−the set of Borel probability measures on H. For given $f \in C_b(H)$ and $\lambda \in \mathscr{P}(H)$, we write

$$(f, \lambda) := \int_H f(u)\lambda(\mathrm{d}u).$$

For $\lambda_1, \lambda_2 \in \mathscr{P}(H)$, we set

$$\|\lambda_1 - \lambda_2\|_{\mathcal{L}}^* := \sup_{\substack{f \in \mathrm{Lip}(H) \\ \|f\|_{\mathrm{Lip}(H)} \le 1}} |(f, \lambda_1) - (f, \lambda_2)|,$$

$$\|\lambda_1 - \lambda_2\|_{\mathrm{var}} := \sup_{\Gamma \in \mathscr{B}(H)} |\lambda_1(\Gamma) - \lambda_2(\Gamma)| = \frac{1}{2} \sup_{\substack{f \in C_b(H) \\ \|f\|_\infty \le 1}} |(f, \lambda_1) - (f, \lambda_2)|.$$

The distribution of a random variable ξ is denoted by $\mathscr{D}(\xi)$. For real numbers a and b, we use $a \vee b$ to denote their maximum and $a \wedge b$ for their minimum. We denote by C, C_a, C_ν, etc. positive constants that are not essential to the analysis, with subscripts indicating dependence on specific parameters. For brevity, we use the notation $\lesssim$, $\lesssim_a$, $\lesssim_\nu$, etc., to indicate inequalities valid up to an inessential multiplicative constant, such as C, C_a, or C_ν.

2 Construction of a mixing extension

Under the conditions mentioned in the previous section, the stochastic NS system (1.1) is well-posed and defines a Markov family $(u_t, \mathbb{P}_u)$ parameterized by the initial condition $u \in H$. The following standard energy estimate is proved in the Appendix:

$$\mathbb{E}_u \|u(t)\|^2 \le e^{-at}\|u\|^2 + C_{a,h,\mathcal{B}_0}, \quad t \ge 0, \tag{2.1}$$

with $\mathbb{E}_u$ being the expectation with respect to $\mathbb{P}_u$. Let $S_t(u, \cdot)$ be the flow issued from $u \in H$, and define the associated Markov operators as follows:

$$\mathscr{B}_t f(u) := \int_H f(v) P_t(u, \mathrm{d}v), \qquad \mathscr{B}_t : C_b(H) \to C_b(H),$$

$$\mathscr{B}_t^* \lambda(\Gamma) := \int_H P_t(u, \Gamma)\lambda(\mathrm{d}u), \qquad \mathscr{B}_t^* : \mathscr{P}(H) \to \mathscr{P}(H),$$

where

$$P_t(u, \Gamma) := \mathbb{P}\{S_t(u, \cdot) \in \Gamma\}$$

is the transition function. A measure $\mu \in \mathscr{P}(H)$ is called stationary for the family $(u_t, \mathbb{P}_u)$, if $\mathscr{B}_t^* \mu = \mu$ for any $t > 0$. We now restate the Main Theorem as follows.

Theorem 2.1. *Under the assumptions of the Main Theorem, the family $(u(t), \mathbb{P}_u)$ has a unique stationary measure $\mu \in \mathscr{P}(H)$. Moreover, for any $q > 1$, there is a constant $C_q > 0$ such that*

$$\|\mathscr{B}_t^* \lambda - \mu\|_{\mathcal{L}}^* \le C_q (1 + t)^{-q} \left(1 + \int_H \|u\|^2 \lambda(\mathrm{d}u)\right)$$

for any $\lambda \in \mathscr{P}(H)$.

Scheme of the proof. The proof relies on the coupling method, as described in Section 3.1.3 of [KS12] and [Shi08]. Let us briefly outline the main ideas. Consider a process $(\boldsymbol{u}_t, \mathbb{P}_{\boldsymbol{u}})$ in $H \times H$, and let π_1 and π_2 be the projections from $H \times H$ to the first and second component. The process $(\boldsymbol{u}_t, \mathbb{P}_{\boldsymbol{u}})$ is said to be an extension of $(u_t, \mathbb{P}_u)$, if for any $\boldsymbol{u} = (u, u') \in H \times H$, the laws under $\mathbb{P}_{\boldsymbol{u}}$ of processes $\{\pi_1 \boldsymbol{u}_t\}_{t\geq 0}$ and $\{\pi_2 \boldsymbol{u}_t\}_{t\geq 0}$ coincide with those of $\{u_t\}_{t\geq 0}$ under $\mathbb{P}_u$ and $\mathbb{P}_{u'}$, respectively. The key idea of the coupling approach is to construct an extension $(\boldsymbol{u}_t, \mathbb{P}_{\boldsymbol{u}})$ that possesses recurrence and polynomial squeezing properties, as described in the following theorem.

Theorem 2.2. *Under the assumptions of the Main Theorem, for any $q > 1$, there are parameters $d, T > 0$, an extension $(\boldsymbol{u}_t, \mathbb{P}_{\boldsymbol{u}})$ of $(u_t, \mathbb{P}_u)$, and a stopping time σ such that $(\boldsymbol{u}_{kT}, \mathbb{P}_{\boldsymbol{u}})$ is a Markov process and the following properties hold.*

1. **(Recurrence)** *There are constants $\delta, C > 0$ such that*
$$\mathbb{E}_{\boldsymbol{u}} \exp(\delta \tau_d) \leq C\left(1 + \|u\|^2 + \|u'\|^2\right) \tag{2.2}$$
for any $\boldsymbol{u} \in H \times H$, where
$$\tau_d := \inf\left\{k \geq 0 \,|\, \boldsymbol{u}_{kT} \in \overline{B}_H(0,d) \times \overline{B}_H(0,d)\right\}. \tag{2.3}$$

2. **(Polynomial squeezing)** *There are constants $\delta_1, c > 0$, which are independent of q, such that*
$$\|\tilde{u}(t) - \tilde{u}'(t)\|^2 \leq C_q e^{-ct}\|u - u'\|^2 \qquad \text{for } 0 \leq t \leq \sigma, \tag{2.4}$$
$$\mathbb{P}_{\boldsymbol{u}}\{\sigma = \infty\} \geq \delta_1, \tag{2.5}$$
$$\mathbb{E}_{\boldsymbol{u}}\left(\mathbb{I}_{\{\sigma<\infty\}}\sigma^q\right) \leq C_q, \tag{2.6}$$
$$\mathbb{E}_{\boldsymbol{u}}\left(\mathbb{I}_{\{\sigma<\infty\}}\left(\|\tilde{u}(\sigma)\|^{2q} + \|\tilde{u}'(\sigma)\|^{2q}\right)\right) \leq C_q \tag{2.7}$$
for any $\boldsymbol{u} \in \overline{B}_H(0,d) \times \overline{B}_H(0,d)$, where $\tilde{u}_t = \pi_1 \boldsymbol{u}_t$ and $\tilde{u}'_t = \pi_2 \boldsymbol{u}_t$.

An extension is said to be mixing if it satisfies the properties in the above theorem. According to Theorem 3.1.7 in [KS12], the existence of a mixing extension ensures the existence of a unique stationary measure, as well as mixing for the semigroup $\mathscr{B}^*_{kT}$. This, in turn, implies mixing for $\mathscr{B}^*_t$ by virtue of (2.1). The criterion given in Theorem 3.1.7 in [KS12] guarantees the exponential mixing; the necessary adaptations to the polynomial mixing case are done in Theorem 1.2 in [Gao24].

Construction. Let us outline the construction of a mixing extension $(\boldsymbol{u}_t, \mathbb{P}_{\boldsymbol{u}})$, following the strategies developed in [Mar14] and [NZ24]. For any $u, u' \in H$, let $\{u(t)\}_{t\geq 0}$ and $\{u'(t)\}_{t\geq 0}$ be the solutions of (1.1) with initial conditions u and u'. Let the process $\{v(t)\}_{t\geq 0}$ be the solution of the following auxiliary problem:
$$\begin{cases} \partial_t v + \Pi(v\cdot\nabla)v - \nu\Delta v + av \\ \qquad\qquad + \mathrm{P}_N[\Pi(u\cdot\nabla)u - \Pi(v\cdot\nabla)v - \nu\Delta(u-v)] = h + \eta, \\ v|_{t=0} = u', \end{cases} \tag{2.8}$$

where P_N is the orthogonal projection in H onto the space spanned by $\{e_1, \dots, e_N\}$, with $N \geq 1$ to be specified later, Π is the Leray projector, and we used the fact that $\Pi\Delta = \Delta$ on $\mathbb{R}^2$.

Let us fix a time step $T > 0$ to be specified later, and denote by $\lambda_T(u, u')$ and $\lambda'_T(u, u')$ the distributions of processes $\{v(t)\}_{t\in[0,T]}$ and $\{u'(t)\}_{t\in[0,T]}$. By Theorem 1.2.28 in [KS12], there is a maximal coupling $(\mathcal{V}_T(u, u'), \mathcal{V}'_T(u, u'))$ for the pair of measures $(\lambda_T(u, u'), \lambda'_T(u, u'))$ defined on some probability space $(\tilde{\Omega}, \tilde{\mathscr{F}}, \tilde{\mathbb{P}})$. That is, we have (see Definition 1.2.21 in [KS12])

$$\mathbb{P}\{\mathcal{V}_T(u, u') \neq \mathcal{V}'_T(u, u')\} = \|\lambda_T(u, u') - \lambda'_T(u, u')\|_{var},$$

and conditioned on the event $\{\mathcal{V}_T(u, u') \neq \mathcal{V}'_T(u, u')\}$, the random variables $\mathcal{V}_T(u, u')$ and $\mathcal{V}'_T(u, u')$ are independent. Let $\{\tilde{v}(t)\}_{t\in[0,T]}$ and $\{\tilde{u}'(t)\}_{t\in[0,T]}$ be the flows of this maximal coupling. Then, the process $\{\tilde{v}(t)\}_{t\in[0,T]}$ is the solution of

$$\begin{cases} \partial_t \tilde{v} + \Pi(\tilde{v} \cdot \nabla)\tilde{v} - \nu\Delta\tilde{v} + a\tilde{v} + \mathrm{P}_N[\nu\Delta\tilde{v} - \Pi(\tilde{v} \cdot \nabla)\tilde{v}] = h + \Lambda, \\ \tilde{v}|_{t=0} = u', \end{cases} \tag{2.9}$$

where the distribution of the process $\left\{\int_0^t \Lambda(s)\mathrm{d}s\right\}_{t\in[0,T]}$ is equal to that of

$$\left\{\int_0^t (\eta(s) - \mathrm{P}_N\Pi(u \cdot \nabla)u + \nu\mathrm{P}_N\Delta u)\,\mathrm{d}s\right\}_{t\in[0,T]}$$

and η is defined by (1.3). Let $\{\tilde{u}(t)\}_{t\in[0,T]}$ be the solution of

$$\begin{cases} \partial_t \tilde{u} + \Pi(\tilde{u} \cdot \nabla)\tilde{u} - \nu\Delta\tilde{u} + a\tilde{u} + \mathrm{P}_N\left[\nu\Delta\tilde{u} - \Pi(\tilde{u} \cdot \nabla)\tilde{u}\right] = h + \Lambda, \\ \tilde{u}|_{t=0} = u. \end{cases} \tag{2.10}$$

Note that, by the uniqueness in law for the equation (2.10), we have

$$\mathscr{D}(\{\tilde{u}(t)\}_{t\in[0,T]}) = \mathscr{D}(\{u(t)\}_{t\in[0,T]}).$$

For any $u, u' \in H$, $\omega \in \tilde{\Omega}$, and $t \in [0, T]$, let us define

$$\mathcal{R}_t(u, u', \omega) := \tilde{u}_t, \quad \mathcal{R}'_t(u, u', \omega) := \tilde{u}'_t.$$

Consider a sequence of independent copies $\{(\Omega^k, \mathscr{F}^k, \mathbb{P}^k)\}_{k\geq 0}$ of the probability space $(\tilde{\Omega}, \tilde{\mathscr{F}}, \tilde{\mathbb{P}})$, and let $(\Omega, \mathscr{F}, \mathbb{P})$ denote their direct product. For any $u, u' \in H$ and $\omega = (\omega^1, \omega^2, \dots) \in \Omega$, let $\tilde{u}_0 = u$, $\tilde{u}'_0 = u'$, and

$$\tilde{u}_t(\omega) := \mathcal{R}_s(\tilde{u}_{kT}(\omega), \tilde{u}'_{kT}(\omega), \omega^k), \quad \tilde{u}'_t(\omega) := \mathcal{R}'_s(\tilde{u}_{kT}(\omega), \tilde{u}'_{kT}(\omega), \omega^k),$$

where $t = s + kT$, $s \in [0, T)$, and $k \geq 1$. Finally, we introduce the pair

$$\boldsymbol{u}_t := (\tilde{u}_t, \tilde{u}'_t).$$

This construction ensures that $(\boldsymbol{u}_t, \mathbb{P}_{\boldsymbol{u}})$ is an extension for $(u_t, \mathbb{P}_u)$. For suitable choices of the parameters N and T, we will show in Section 5 that the process $(\boldsymbol{u}_t, \mathbb{P}_{\boldsymbol{u}})$ is a mixing extension for $(u_t, \mathbb{P}_u)$. To prepare for this, in Sections 3 and 4, we establish the necessary groundwork by studying some growth estimates for energy functionals and stability properties of the stochastic NS system (1.1).

3 Growth estimates for solutions

The main result of this section is Proposition 3.5, which establishes a growth estimate for the weighted energy functional defined as follows:

$$\begin{aligned}\mathcal{E}^u_\psi(t) := \|u(t)\|^2 + \|u(t)\|^6 + \int_0^t \left(\|u(s)\|^2_{H^1} + \|u(s)\|^4\|\nabla u(s)\|^2 + \|u(s)\|^6\right) \mathrm{d}s \\ + \|\psi(t)u(t+1)\|^2 + \int_0^t \left(\|\psi(s)\nabla u(s+1)\|^2 + \|\psi(s)u(s+1)\|^2\right) \mathrm{d}s \\ + \|w(t+1)\|^2 + \int_0^t \|w(s+1)\|^2_{H^1} \mathrm{d}s \\ + \|\psi(t)w(t+1)\|^2 + \int_0^t \left(\|\psi(s)\nabla w(s+1)\|^2 + \|\psi(s)w(s+1)\|^2\right) \mathrm{d}s,\end{aligned} \tag{3.1}$$

where $u(t)$ is the solution of the stochastic NS system (1.1) issued from u_0 and $w := \operatorname{curl} u$ is the corresponding vorticity. The space-time weight function ψ is given by

$$\psi(t,x) := \varphi(x)\left(1 - \exp\left(-\frac{t}{\varphi(x)}\right)\right) \tag{3.2}$$

with $\varphi(x) := \sqrt{|x|^2+1}$. The weight function ψ possesses the following properties, which will be used throughout the paper:

- **(i)** $0 < \psi(t,x) < \varphi(x)$ and $\psi(0,x) = 0$ for any $t > 0$ and $x \in \mathbb{R}^2$;
- **(ii)** the partial derivatives of ψ of order ≥ 1 are bounded;
- **(iii)** as $t, |x| \to +\infty$, there holds $\psi(t,x) \to +\infty$;
- **(iv)** $\psi(t), t \geq 2$ belongs to the Muckenhoupt A_2 class and $\sup_{t\geq 2}[\psi(t)]_{A_2} < \infty$.

The proof of the properties (i)-(iii) is straightforward, cf. Section 2.1 in [NZ24], and (iv) is established in Lemma B.3. To estimate the energy $\mathcal{E}^u_\psi(t)$, we break it down into several components, each of which is estimated in the following four subsections. In what follows, we always assume that the hypotheses of the Main Theorem are satisfied.

3.1 L^2-estimate for the velocity field

Let us begin by estimating the L^2-energy of the velocity field raised to the power $p \geq 1$:

$$\mathcal{E}^u_p(t) := \|u(t)\|^{2p} + \int_0^t \left(\|u(s)\|^{2p-2}\|\nabla u(s)\|^2 + \|u(s)\|^{2p}\right) \mathrm{d}s. \tag{3.3}$$

Proposition 3.1. *There exist constants κ_p and $\mathcal{C}_p$ depending on $a, \nu, h, \mathcal{B}_0, p$ such that the following estimate holds:*

$$\mathbb{P}\left\{\sup_{t\geq 0}\left(\mathcal{E}_p^u(t)-\kappa_p t-\mathcal{C}_p\|u_0\|^{2p}\right)\geq\rho\right\}\leq C_{a,\nu,h,\mathcal{B}_0,p,q}\frac{\mathbb{E}\|u_0\|^{(2p-1)q}+1}{\rho^{\frac{q}{2}-1}}$$

for any $q, \rho > 2$.

Proof. By the Itô formula and the cancellation property of the convection term, we have

$$\begin{aligned}\mathrm{d}\|u\|^{2p}=p\|u\|^{2(p-1)}&\left(2\langle u,\nu\Delta u-au+h\rangle\mathrm{d}t+2\sum_{j=1}^{\infty}b_j\langle u,e_j\rangle\mathrm{d}\beta_j+\mathcal{B}_0\mathrm{d}t\right)\\&+2p(p-1)\|u\|^{2(p-2)}\sum_{j=1}^{\infty}b_j^2\langle u,e_j\rangle^2\mathrm{d}t.\end{aligned}$$

Integrating by parts, we derive

$$\begin{aligned}\|u(t)\|^{2p}+\int_0^t\left(2p\nu\|\nabla u\|^2\|u\|^{2(p-1)}+2ap\|u\|^{2p}\right)\mathrm{d}s&\\\leq\|u_0\|^{2p}+\int_0^t p\|u\|^{2(p-1)}\left(2\langle u,h\rangle+(2p-1)\mathcal{B}_0\right)\mathrm{d}s+M_p(t),&\quad(3.4)\end{aligned}$$

where

$$M_p(t):=2p\int_0^t\|u\|^{2(p-1)}\sum_{j=1}^{\infty}b_j\langle u,e_j\rangle\mathrm{d}\beta_j.$$

Then, by the Young inequality,

$$\mathcal{E}_p^u(t)-C_{a,\nu,p}\|u_0\|^{2p}\leq C_{a,\nu,h,\mathcal{B}_0,p}t+C_{a,\nu,p}M_p(t),\tag{3.5}$$

which implies that

$$\mathcal{E}_p^u(t)-\kappa_p t-C_{a,\nu,p}\|u_0\|^{2p}\leq C_{a,\nu,p}\left(M_p(t)-t\right)$$

for some positive constant κ_p. Therefore,

$$\begin{aligned}\left\{\sup_{t\geq 0}\left(\mathcal{E}_p^u(t)-\kappa_p t-C_{a,\nu,p}\|u_0\|^{2p}\right)\geq\rho\right\}&\subset\left\{\sup_{t\geq 0}\left(M_p(t)-t\right)\geq\frac{\rho}{C_{a,\nu,p}}\right\}\\&=\bigcup_{n=0}^{\infty}\left\{\sup_{t\in[n,n+1)}\left(M_p(t)-t\right)\geq\frac{\rho}{C_{a,\nu,p}}\right\}\\&\subset\bigcup_{n=0}^{\infty}\left\{\sup_{t<n+1}|M_p(t)|\geq\frac{\rho}{C_{a,\nu,p}}+n\right\}.\end{aligned}$$

An application of the Chebyshev and Burkholder–Davis–Gundy inequalities further yields

$$\begin{aligned}\mathbb{P}\left\{\sup_{t\geq 0}\left(\mathcal{E}_p^u(t)-\kappa_p t-C_{a,\nu,p}\|u_0\|^{2p}\right)\geq\rho\right\}&\\ \leq\sum_{n=0}^{\infty}\mathbb{P}\left\{\sup_{t<n+1}|M_p(t)|\geq\frac{\rho}{C_{a,\nu,p}}+n\right\}&\\ \leq C_{a,\nu,p,q}\sum_{n=0}^{\infty}\frac{\mathbb{E}\sup_{t<n+1}|M_p(t)|^q}{(\rho+n)^q}&\\ \leq C_{a,\nu,p,q}\sum_{n=0}^{\infty}\frac{\mathbb{E}\left(\langle M_p\rangle(n+1)\right)^{\frac{q}{2}}}{(\rho+n)^q},&\qquad(3.6)\end{aligned}$$

where

$$\langle M_p\rangle(t):=4p^2\int_0^t\|u\|^{4(p-1)}\sum_{j=1}^{\infty}b_j^2\langle u,e_j\rangle^2\mathrm{d}s$$

is the quadratic variation of the martingale M_p. Applying the Hölder inequality, we find that

$$\langle M_p\rangle^{\frac{q}{2}}(t)\leq(2p)^q\left(\int_0^t\mathcal{B}_0\|u\|^{4p-2}\mathrm{d}s\right)^{\frac{q}{2}}\leq(2p)^q\mathcal{B}_0^{\frac{q}{2}}t^{\frac{q}{2}-1}\int_0^t\|u\|^{(2p-1)q}\mathrm{d}s.\quad(3.7)$$

To bound the term on the right-hand side of this inequality, we take the expectation in (3.5) with the parameter p replaced by $(p-\frac{1}{2})q$, yielding

$$\mathbb{E}\int_0^t\|u\|^{(2p-1)q}\mathrm{d}s\leq C_{a,\nu,p,q}\mathbb{E}\|u_0\|^{(2p-1)q}+C_{a,\nu,h,\mathcal{B}_0,p,q}t.\qquad(3.8)$$

Combining the estimates (3.6)–(3.8), we arrive at

$$\begin{aligned}\mathbb{P}\left\{\sup_{t\geq 0}\left(\mathcal{E}_p^u(t)-\kappa_p t-C_{a,\nu,p}\|u_0\|^{2p}\right)\geq\rho\right\}&\\ \leq C_{a,\nu,h,\mathcal{B}_0,p,q}\sum_{n=0}^{\infty}\frac{\mathbb{E}\|u_0\|^{(2p-1)q}+1}{(\rho+n)^{\frac{q}{2}}}.&\end{aligned}$$

As

$$\sum_{n=0}^{\infty}\frac{1}{(\rho+n)^{\frac{q}{2}}}\leq\int_{\rho-1}^{\infty}\frac{1}{x^{\frac{q}{2}}}\mathrm{d}x=\frac{2}{q-2}\frac{1}{(\rho-1)^{\frac{q}{2}-1}}\lesssim_q\frac{1}{\rho^{\frac{q}{2}-1}}$$

for $q>2$, the proof is complete. □

3.2 L^2-estimate for the vorticity

Next, we turn to the L^2-energy of the vorticity $w = \operatorname{curl} u$ to the power $p \geq 1$:

$$\mathcal{E}^u_{1,p}(t) := \|w(t+1)\|^{2p} + \int_0^t \left(\|w(s+1)\|^{2p-2} \|\nabla w(s+1)\|^2 + \|w(s+1)\|^{2p} \right) \mathrm{d}s.$$

Let us note that

$$\mathcal{B}_1 := \sum_{j=1}^{\infty} b_j^2 \|e_j\|^2_{H^1} < \infty,$$

due to the growth of φ and the condition $\mathcal{B}_\varphi < \infty$.

Proposition 3.2. *There exist constants $\kappa_{1,p}$ and $\mathcal{C}_{1,p}$ depending on $a, \nu, h, \mathcal{B}_1, p$ such that*

$$\mathbb{P}\left\{ \sup_{t \geq 0} \left(\mathcal{E}^u_{1,p}(t) - \kappa_{1,p} t - \mathcal{C}_{1,p} \|w(1)\|^{2p} \right) \geq \rho \right\} \leq C_{a,\nu,h,\mathcal{B}_1,p,q} \frac{\mathbb{E}\|u_0\|^{4(pq+2)} + 1}{\rho^{\frac{q}{2}-1}}$$

for any $q, \rho > 2$.

Proof. The proof is similar to that of Proposition 3.1. The equation for w is given by

$$\partial_t w + u \cdot \nabla w - \nu \Delta w + a w = \operatorname{curl} h + \operatorname{curl} \eta. \tag{3.9}$$

By the Itô formula and the cancellation property of the convection term,

$$\begin{aligned} \mathrm{d}\|w\|^{2p} &= 2p\|w\|^{2(p-1)} \langle w, \nu \Delta w - a w + \operatorname{curl} h \rangle \mathrm{d}t \\ &\quad + p\|w\|^{2(p-1)} \left(2 \sum_{j=1}^{\infty} b_j \langle w, \operatorname{curl} e_j \rangle \mathrm{d}\beta_j + \sum_{j=1}^{\infty} b_j^2 \| \operatorname{curl} e_j \|^2 \mathrm{d}t \right) \\ &\quad + 2p(p-1)\|w\|^{2(p-2)} \sum_{j=1}^{\infty} b_j^2 \langle w, \operatorname{curl} e_j \rangle^2 \mathrm{d}t. \end{aligned}$$

Integrating this equality from 1 to $t+1$, we derive

$$\begin{aligned} &\|w(t+1)\|^{2p} - \|w(1)\|^{2p} + \int_1^{t+1} \left(2p\nu \|w\|^{2(p-1)} \|\nabla w\|^2 + 2ap\|w\|^{2p} \right) \mathrm{d}s \\ &\quad \leq \int_1^{t+1} p\|w\|^{2(p-1)} \left(2\langle w, \operatorname{curl} h \rangle + (2p-1)\mathcal{B}_1 \right) \mathrm{d}s + M_{1,p}(t), \end{aligned}$$

where

$$M_{1,p}(t) := 2p \int_1^{t+1} \|w\|^{2(p-1)} \sum_{j=1}^{\infty} b_j \langle w, \operatorname{curl} e_j \rangle \mathrm{d}\beta_j.$$

Applying the Young inequality, we find a positive constant $\kappa_{1,p}$ depending on $a, \nu, h, \mathcal{B}_1, p$ such that

$$\mathcal{E}^u_{1,p}(t) - \kappa_{1,p} t - C_{a,\nu,p}\|w(1)\|^{2p} \le C_{a,\nu,p}\left(M_{1,p}(t) - t\right). \tag{3.10}$$

Repeating the arguments of the proof of (3.6), we obtain

$$\begin{aligned}\mathbb{P}\left\{\sup_{t\ge 0}\left(\mathcal{E}^u_{1,p}(t) - \kappa_{1,p}t - C_{a,\nu,p}\|w(1)\|^{2p}\right) \ge \rho\right\} \\ \le C_{a,\nu,p,q}\sum_{n=0}^{\infty}\frac{\mathbb{E}\left(\langle M_{1,p}\rangle(n+1)\right)^{\frac{q}{2}}}{(\rho+n)^q},\end{aligned} \tag{3.11}$$

where

$$\langle M_{1,p}\rangle(t) := 4p^2\int_1^{t+1}\|w\|^{4(p-1)}\sum_{j=1}^{\infty} b_j^2\langle w, \operatorname{curl} e_j\rangle^2 \mathrm{d}s$$

is the quadratic variation of $M_{1,p}(t)$. The Hölder inequality implies that

$$\begin{aligned}\langle M_{1,p}\rangle^{\frac{q}{2}}(t) &\le (2p)^q\left(\int_1^{t+1}\mathcal{B}_1\|w\|^{4p-2}\mathrm{d}s\right)^{\frac{q}{2}} \\ &\le (2p)^q\mathcal{B}_1^{\frac{q}{2}}t^{\frac{q}{2}-1}\int_1^{t+1}\|w\|^{(2p-1)q}\mathrm{d}s.\end{aligned} \tag{3.12}$$

From the inequality (3.10) with p replaced by $(p-\frac{1}{2})q$ and Lemma A.1 it follows that

$$\begin{aligned}\mathbb{E}\int_1^t\|w\|^{(2p-1)q}\mathrm{d}s &\le C_{a,\nu,p,q}\mathbb{E}\|w(1)\|^{(2p-1)q} + C_{a,\nu,h,\mathcal{B}_1,p,q}t \\ &\le C_{a,\nu,h,\mathcal{B}_1,p,q}\left(\mathbb{E}\|u_0\|^{4(pq+2)} + t\right), \quad t \ge 1.\end{aligned} \tag{3.13}$$

Combining the estimates (3.11)–(3.13), we derive

$$\begin{aligned}\mathbb{P}\left\{\sup_{t\ge 0}\left(\mathcal{E}^u_{1,p}(t) - \kappa_{1,p}t - C_{a,\nu,p}\|w(1)\|^{2p}\right) \ge \rho\right\} \\ \le C_{a,\nu,h,\mathcal{B}_1,p,q}\sum_{n=0}^{\infty}\frac{\mathbb{E}\|u_0\|^{4(pq+2)}+1}{(\rho+n)^{\frac{q}{2}}} \\ \le C_{a,\nu,h,\mathcal{B}_1,p,q}\frac{\mathbb{E}\|u_0\|^{4(pq+2)}+1}{\rho^{\frac{q}{2}-1}},\end{aligned}$$

which completes the proof. □

3.3 Weighted estimate for the velocity field

Here we establish a growth estimate for the weighted energy of the velocity field:

$$\tilde{\mathcal{E}}^u_\psi(t) := \|\psi(t)u(t+1)\|^2 + \int_0^t\left(\|\psi(s)\nabla u(s+1)\|^2 + \|\psi(s)u(s+1)\|^2\right)\mathrm{d}s.$$

Proposition 3.3. *There exist constants $\tilde{\kappa}, \tilde{\gamma}, \tilde{\mathcal{C}}$ depending on $a, \nu, h, \mathcal{B}_0, \mathcal{B}_\varphi$ such that for any $q, \rho > 2$,*

$$\mathbb{P}\left\{\sup_{t\geq 0}\left(\tilde{\mathcal{E}}^u_{\tilde{\psi}}(t) - \tilde{\kappa}t - \tilde{\mathcal{C}}\left(1+\|u_0\|^6\right)\right) \geq \rho\right\} \leq C_{a,\nu,h,\mathcal{B}_0,q}\left(\mathbb{E}\|u_0\|^{5q}+1\right)\left(e^{-\tilde{\gamma}\rho} + \frac{1}{\rho^{\frac{q}{2}-1}}\right).$$

Proof. To simplify the notation, let us write $\tilde{\psi}(t+1) := \psi(t)$. By the Itô formula,

$$\begin{aligned}\mathrm{d}\|\tilde{\psi}u\|^2 &= 2\langle \partial_t\tilde{\psi}u, \tilde{\psi}u\rangle + 2\langle \tilde{\psi}u, \tilde{\psi}(-(u\cdot\nabla)u + \nu\Delta u - au + h - \nabla p)\rangle \mathrm{d}t \\ &\quad + \mathrm{d}M_{\tilde{\psi}}(t) + \sum_{j=1}^{\infty} b_j^2\|\tilde{\psi}e_j\|^2\mathrm{d}t, \end{aligned} \tag{3.14}$$

where

$$M_{\tilde{\psi}}(t) := 2\int_1^{t+1}\sum_{j=1}^{\infty} b_j\langle \tilde{\psi}u, \tilde{\psi}e_j\rangle \mathrm{d}\beta_j.$$

For the nonlinear convection term, we use the property (ii) of the weight function ψ and the Ladyzhenskaya inequality

$$\|f\|_{L^4} \lesssim \|f\|^{\frac{1}{2}}\|\nabla f\|^{\frac{1}{2}} \tag{3.15}$$

to derive

$$\begin{aligned} -\langle \tilde{\psi}u, \tilde{\psi}(u\cdot\nabla)u\rangle &= \langle \nabla\tilde{\psi}\cdot u, \tilde{\psi}|u|^2\rangle \\ &\leq C\|\tilde{\psi}u\|_{L^4}\|u\|_{L^4}\|u\| \\ &\leq C\|\tilde{\psi}u\|^{\frac{1}{2}}\|\tilde{\psi}\nabla u\|^{\frac{1}{2}}\|u\|^{\frac{3}{2}}\|\nabla u\|^{\frac{1}{2}} + C\|\tilde{\psi}u\|^{\frac{1}{2}}\|u\|^2\|\nabla u\|^{\frac{1}{2}} \\ &\leq \delta\|\tilde{\psi}u\|^2 + \delta\|\tilde{\psi}\nabla u\|^2 + C\|\nabla u\|^2 + C\left(1+\|u\|^6\right), \end{aligned} \tag{3.16}$$

where $\delta > 0$ is a small parameter to be determined. As for the pressure term, using again the property (ii) of the weight function ψ,

$$\langle \tilde{\psi}u, \tilde{\psi}\nabla p\rangle = -2\langle \tilde{\psi}u, \nabla\tilde{\psi}p\rangle \leq C\|\tilde{\psi}u\|\|p\|. \tag{3.17}$$

Moreover, since p satisfies

$$-\Delta p = \mathrm{div}((u\cdot\nabla)u) = \sum_{k,j=1}^{2}\partial_k\partial_j(u_ku_j),$$

we have

$$p = -\sum_{k,j=1}^{2}\mathcal{R}_k\mathcal{R}_j(u_ku_j),$$

where

$$\mathcal{R}_j f(x) := \mathcal{F}^{-1}\left(-i\frac{\xi_j}{|\xi|}\mathcal{F}f(\xi)\right)(x) \tag{3.18}$$

is the Riesz transform and $\mathcal{F}$ is the Fourier transform. By the L^2-boundedness of $\mathcal{R}_j$ and the Ladyzhenskaya inequality (3.15), we have

$$\|p\| \lesssim \sum_{i,j=1}^{2} \|u_i u_j\| \lesssim \|u\|_{L^4}^2 \lesssim \|u\|\|\nabla u\|.$$

Combining this with (3.17) and the Young inequality, we obtain

$$\langle \tilde{\psi} u, \tilde{\psi} \nabla p \rangle \leq C\|\tilde{\psi} u\|\|u\|\|\nabla u\| \leq \delta \|\tilde{\psi} u\|^2 + C\|u\|^2\|\nabla u\|^2. \tag{3.19}$$

Notice that

$$\sum_{j=1}^{\infty} b_j^2 \|\tilde{\psi}(s) e_j\|^2 \leq \sum_{j=1}^{\infty} b_j^2 \|\varphi e_j\|^2 \leq \mathcal{B}_\varphi, \qquad s \geq 1,$$

due to the property (i) of the weight function ψ. Then, choosing $\delta := (a \wedge \nu)/4$, plugging the estimates (3.16) and (3.19) into the equality (3.14), integrating the resulting inequality from 1 to $t+1$, and using Cauchy–Schwartz inequality, we obtain

$$\begin{aligned}
\|\psi(t)u(t+1)\|^2 + \int_0^t \left(\nu\|\psi(s)\nabla u(s+1)\|^2 + a\|\psi(s)u(s+1)\|^2\right) \mathrm{d}s \\
\leq C_{a,\nu} \int_1^{t+1} \left(\|u\|^2\|\nabla u\|^2 + \|\nabla u\|^2 + \|u\|^6\right) \mathrm{d}s \\
+ C_{a,\nu,h,\mathcal{B}_\varphi} t + M_{\tilde{\psi}}(t).
\end{aligned}$$

As the quadratic variation $\langle M_{\tilde{\psi}} \rangle$ satisfies

$$\begin{aligned}
\langle M_{\tilde{\psi}} \rangle(t) &= 4 \int_1^{t+1} \sum_{j=1}^{\infty} b_j^2 \langle \tilde{\psi} u, \tilde{\psi} e_j \rangle^2 \mathrm{d}s \leq 4\mathcal{B}_\varphi \int_1^{t+1} \|\tilde{\psi} u\|^2 \mathrm{d}s \\
&= 4\mathcal{B}_\varphi \int_0^t \|\psi(s)u(s+1)\|^2 \mathrm{d}s,
\end{aligned}$$

by setting $\tilde{\gamma}' := \frac{a}{4\mathcal{B}_\varphi}$, we get

$$\begin{aligned}
\tilde{\mathcal{E}}_\psi^u(t) \leq C_{a,\nu} \left(\mathcal{E}_1^u(t+1) + \mathcal{E}_2^u(t+1) + \mathcal{E}_3^u(t+1)\right) \\
+ C_{a,\nu,h,\mathcal{B}_\varphi} t + C_{a,\nu} \left(M_{\tilde{\psi}}(t) - \frac{\tilde{\gamma}'}{2} \langle M_{\tilde{\psi}} \rangle(t)\right).
\end{aligned}$$

For $p \geq 1$, let $(\kappa_p, \mathcal{C}_p)$ be the constants in Proposition 3.1. Define

$$\tilde{\kappa} := C_{a,\nu,h,\mathcal{B}_\varphi} + C_{a,\nu} \left(\kappa_1 + \kappa_2 + \kappa_3\right),$$

and let $\tilde{\mathcal{C}}$ be such that

$$C_{a,\nu} \left(\kappa_1 + \kappa_2 + \kappa_3 + \mathcal{C}_1\|u_0\|^2 + \mathcal{C}_2\|u_0\|^4 + \mathcal{C}_3\|u_0\|^6\right) \leq \tilde{\mathcal{C}} \left(1 + \|u_0\|^6\right).$$

Then,

$$\begin{aligned}\tilde{\mathcal{E}}^u_\psi(t) - \tilde{\kappa}t - \tilde{\mathcal{C}}\left(1+\|u_0\|^6\right) \le\ & C_{a,\nu}\left(\mathcal{E}^u_1(t+1) - \kappa_1(t+1) - \mathcal{C}_1\|u_0\|^2\right)\\ &+ C_{a,\nu}\left(\mathcal{E}^u_2(t+1) - \kappa_2(t+1) - \mathcal{C}_2\|u_0\|^4\right)\\ &+ C_{a,\nu}\left(\mathcal{E}^u_3(t+1) - \kappa_3(t+1) - \mathcal{C}_3\|u_0\|^6\right)\\ &+ C_{a,\nu}\left(M_{\tilde{\psi}}(t) - \frac{\tilde{\gamma}'}{2}\langle M_{\tilde{\psi}}\rangle(t)\right).\end{aligned}$$

Applying the exponential supermartingale estimate (cf. (A.57) in [KS12]) and Propositions 3.1, we conclude that

$$\begin{aligned}&\mathbb{P}\left\{\sup_{t\ge 0}\left(\tilde{\mathcal{E}}^u_\psi(t) - \tilde{\kappa}t - \tilde{\mathcal{C}}\left(1+\|u_0\|^6\right)\right) \ge \rho\right\}\\ &\qquad\le \mathbb{P}\left\{\sup_{t\ge 0}\left(\mathcal{E}^u_1(t) - \kappa_1 t - \mathcal{C}_1\|u_0\|^2\right) \ge \frac{\rho}{4C_{a,\nu}}\right\}\\ &\qquad\quad+ \mathbb{P}\left\{\sup_{t\ge 0}\left(\mathcal{E}^u_2(t) - \kappa_2 t - \mathcal{C}_2\|u_0\|^4\right) \ge \frac{\rho}{4C_{a,\nu}}\right\}\\ &\qquad\quad+ \mathbb{P}\left\{\sup_{t\ge 0}\left(\mathcal{E}^u_3(t) - \kappa_3 t - \mathcal{C}_3\|u_0\|^6\right) \ge \frac{\rho}{4C_{a,\nu}}\right\}\\ &\qquad\quad+ \mathbb{P}\left\{\sup_{t\ge 0}\left(M_{\tilde{\psi}}(t) - \frac{\tilde{\gamma}'}{2}\langle M_{\tilde{\psi}}\rangle(t)\right) \ge \frac{\rho}{4C_{a,\nu}}\right\}\\ &\qquad\le C_{a,\nu,h,\mathcal{B}_0,q}\left(\mathbb{E}\|u_0\|^{5q}+1\right)\left(e^{-\tilde{\gamma}\rho} + \frac{1}{\rho^{\frac{q}{2}-1}}\right),\end{aligned}$$

where $\tilde{\gamma} := \frac{\tilde{\gamma}'}{4C_{a,\nu}}$. □

3.4 Weighted estimate for the vorticity

Our next goal is to derive a growth estimate for the weighted energy of the vorticity:

$$\tilde{\mathcal{E}}^u_{1,\psi}(t) := \|\psi(t)w(t+1)\|^2 + \int_0^t \left(\|\psi(s)\nabla w(s+1)\|^2 + \|\psi(s)w(s+1)\|^2\right)\mathrm{d}s.$$

Proposition 3.4. *There exist constants $\tilde{\kappa}_1, \tilde{\gamma}_1, \tilde{\mathcal{C}}_1$ depending on $a, \nu, h, \mathcal{B}_1, \mathcal{B}_\varphi$ such that for any $q, \rho > 2$,*

$$\begin{aligned}&\mathbb{P}\left\{\sup_{t\ge 0}\left(\tilde{\mathcal{E}}^u_{1,\psi}(t) - \tilde{\kappa}_1 t - \tilde{\mathcal{C}}_1(1+\|u_0\|^6 + \|w(1)\|^4)\right) \ge \rho\right\}\\ &\qquad\qquad\le C_{a,\nu,h,\mathcal{B}_1,q}\left(\mathbb{E}\|u_0\|^{8(q+1)}+1\right)\left(e^{-\tilde{\gamma}_1\rho} + \frac{1}{\rho^{\frac{q}{2}-1}}\right).\end{aligned}$$

Proof. Again, we set $\tilde{\psi}(t+1) := \psi(t)$ and proceed as in the proof of Proposition 3.3. By the Itô formula and the equation (3.9),

$$\begin{aligned}\mathrm{d}\|\tilde{\psi}w\|^2 &= 2\langle \partial_t\tilde{\psi}w, \tilde{\psi}w\rangle + 2\langle \tilde{\psi}w, \tilde{\psi}(-u\cdot\nabla w + \nu\Delta w - aw + \operatorname{curl} h)\rangle \mathrm{d}t \\ &\quad + \mathrm{d}M_{1,\tilde{\psi}}(t) + \sum_{j=1}^{\infty} b_j^2 \|\tilde{\psi}\operatorname{curl} e_j\|^2 \mathrm{d}t, \end{aligned} \tag{3.20}$$

where

$$M_{1,\tilde{\psi}}(t) := 2\int_1^{t+1}\sum_{j=1}^{\infty} b_j \langle \tilde{\psi}w, \tilde{\psi}\operatorname{curl} e_j\rangle \mathrm{d}\beta_j.$$

For the nonlinear convection term, we use the property (ii) of the weight function ψ and the Ladyzhenskaya inequality (3.15) to derive

$$\begin{aligned} -\langle \tilde{\psi}w, \tilde{\psi}u\cdot\nabla w\rangle &= \langle \nabla\tilde{\psi}u, \tilde{\psi}w^2\rangle \\ &\le C\|\tilde{\psi}w\|_{L^4}\|u\|_{L^4}\|w\| \\ &\le C\|\tilde{\psi}w\|^{\frac{1}{2}}\|\nabla(\tilde{\psi}w)\|^{\frac{1}{2}}\|u\|^{\frac{1}{2}}\|w\|^{\frac{3}{2}} \\ &\le \delta\|\tilde{\psi}w\|^2 + \delta\|\nabla(\tilde{\psi}w)\|^2 + C\|w\|^4 + C\|u\|^2\|\nabla u\|^2 \\ &\le \delta\|\tilde{\psi}w\|^2 + \delta\|\tilde{\psi}\nabla w\|^2 + C\left(1 + \|w\|^4 + \|u\|^2\|\nabla u\|^2\right), \end{aligned} \tag{3.21}$$

where $\delta > 0$ is a small parameter to be determined. Moreover, by the property (i)) of the weight function ψ, we have

$$\langle M_{1,\tilde{\psi}}\rangle(t) = 4\int_1^{t+1}\sum_{j=1}^{\infty} b_j^2\langle \tilde{\psi}w, \tilde{\psi}\operatorname{curl} e_j\rangle^2 \mathrm{d}s \le 4\mathcal{B}_\varphi\int_1^{t+1}\|\tilde{\psi}w\|^2\mathrm{d}s, \tag{3.22}$$

where $\mathcal{B}_\varphi$ is given in (1.6). Taking $\delta := (a\wedge\nu)/4$, plugging (3.21) and (3.22) into (3.20), integrating the resulting inequality from 1 to $t+1$, and using the Cauchy–Schwartz inequality, we derive

$$\begin{aligned} &\|\psi(t)w(t+1)\|^2 + \int_0^t \left(\|\psi(s)\nabla w(s+1)\|^2 + \|\psi(s)w(s+1)\|^2\right)\mathrm{d}s \\ &\qquad \le C_{a,\nu}\int_1^{t+1}\left(\|w\|^4 + \|u\|^2\|\nabla u\|^2\right)\mathrm{d}s + C_{a,\nu,h,\mathcal{B}_\varphi}t \\ &\qquad\quad + C_{a,\nu}\left(M_{1,\tilde{\psi}}(t) - \frac{\tilde{\gamma}_1'}{2}\langle M_{1,\tilde{\psi}}\rangle(t)\right), \end{aligned}$$

where $\tilde{\gamma}_1' := \frac{a}{4\mathcal{B}_\varphi}$. This implies

$$\begin{aligned} \tilde{\mathcal{E}}^u_{1,\psi}(t) &\le C_{a,\nu}\left(\mathcal{E}^u_{1,2}(t) + \mathcal{E}^u_2(t+1)\right) + C_{a,\nu,h,\mathcal{B}_\varphi}t \\ &\quad + C_{a,\nu}\left(M_{1,\tilde{\psi}}(t) - \frac{\tilde{\gamma}_1'}{2}\langle M_{1,\tilde{\psi}}\rangle(t)\right). \end{aligned} \tag{3.23}$$

For $p \geq 1$, let $\kappa_p, \kappa_{1,p}$ be the constants in Propositions 3.1 and 3.2, respectively, and let

$$\tilde{\kappa}_1 := C_{a,\nu}\left(\kappa_{1,2} + \kappa_2\right) + C_{a,\nu,h,\mathcal{B}_\varphi}$$

and choose $\tilde{\mathcal{C}}_1$ as such that

$$C_{a,\nu}\mathcal{C}_{1,2}\|w(1)\|^4 + C_{a,\nu}\mathcal{C}_2\|u_0\|^4 + C_{a,\nu}\kappa_2 \leq \tilde{\mathcal{C}}_1\left(1 + \|u_0\|^6 + \|w(1)\|^4\right).$$

Then, we derive from (3.23) that

$$\begin{aligned}
&\tilde{\mathcal{E}}^u_{1,\psi}(t) - \tilde{\kappa}_1 t - \tilde{\mathcal{C}}_1\left(1 + \|u_0\|^6 + \|w(1)\|^4\right) \\
&\quad \leq C_{a,\nu}\left(M_{1,\tilde{\psi}}(t) - \frac{\tilde{\gamma}_1'}{2}\langle M_{1,\tilde{\psi}}\rangle(t)\right) + C_{a,\nu}\left(\mathcal{E}^u_{1,2}(t) - \kappa_{1,2}t - \mathcal{C}_{1,2}\|w(1)\|^4\right) \\
&\qquad + C_{a,\nu}\left(\mathcal{E}^u_2(t+1) - \kappa_2(t+1) - \mathcal{C}_2|u_0\|^4\right),
\end{aligned}$$

which, by the exponential supermartingale estimate, implies

$$\begin{aligned}
&\mathbb{P}\left\{\sup_{t\geq 0}\left(\tilde{\mathcal{E}}^u_{1,\psi}(t) - \tilde{\kappa}_1 t - \tilde{\mathcal{C}}_1(1 + \|u_0\|^6 + \|w(1)\|^4)\right) \geq \rho\right\} \\
&\qquad \leq \mathbb{P}\left\{\sup_{t\geq 0}\left(\mathcal{E}^u_{1,2}(t) - \kappa_{1,2}t - \mathcal{C}_{1,2}\|w(1)\|^4\right) \geq \frac{\rho}{3C_{a,\nu}}\right\} \\
&\qquad\quad + \mathbb{P}\left\{\sup_{t\geq 0}\left(\mathcal{E}^u_2(t) - \kappa_2 t - \mathcal{C}_2\|u_0\|^4\right) \geq \frac{\rho}{3C_{a,\nu}}\right\} \\
&\qquad\quad + \mathbb{P}\left\{\sup_{t\geq 0}\left(M_{1,\tilde{\psi}}(t) - \frac{\tilde{\gamma}_1'}{2}\langle M_{1,\tilde{\psi}}\rangle(t)\right) \geq \frac{\rho}{3C_{a,\nu}}\right\} \\
&\qquad \leq C_{a,\nu,h,\mathcal{B}_1,q}\left(\mathbb{E}\|u_0\|^{8(q+1)} + 1\right)\left(e^{-\tilde{\gamma}_1\rho} + \frac{1}{\rho^{\frac{q}{2}-1}}\right),
\end{aligned}$$

where $\tilde{\gamma}_1 := \frac{\tilde{\gamma}_1'}{3C_{a,\nu}}$. □

3.5 Estimate for the weighted parabolic energy

Finally, we combine the estimates established in Sections 3.1–3.4 to derive a growth estimate for the weighted parabolic energy $\mathcal{E}^u_\psi(t)$ defined in (3.1).

Proposition 3.5. *There exist constants $\kappa, \gamma, \mathcal{C}$ depending on $a, \nu, h, \mathcal{B}_1, \mathcal{B}_\varphi$ such that for any $q, \rho > 2$,*

$$\begin{aligned}
&\mathbb{P}\left\{\sup_{t\geq 0}\left(\mathcal{E}^u_\psi(t) - \kappa t - \mathcal{C}\left(1 + \|u_0\|^6 + \|w(1)\|^4\right)\right) \geq \rho\right\} \\
&\qquad\qquad \leq C_{a,\nu,h,\mathcal{B}_1,q}\left(\mathbb{E}\|u_0\|^{8(q+1)} + 1\right)\left(e^{-\gamma\rho} + \frac{1}{\rho^{\frac{q}{2}-1}}\right).
\end{aligned}$$

Proof. By definition,

$$\mathcal{E}^u_\psi(t) = \mathcal{E}^u_1(t) + \mathcal{E}^u_3(t) + \tilde{\mathcal{E}}^u_\psi(t) + \mathcal{E}^u_{1,1}(t) + \tilde{\mathcal{E}}^u_{1,\psi}(t),$$

which implies

$$\begin{aligned}&\mathcal{E}^u_\psi(t) - (\kappa_1 + \kappa_3 + \tilde{\kappa} + \kappa_{1,1} + \tilde{\kappa}_1)\, t \\ &\quad = (\mathcal{E}^u_1 - \kappa_1 t) + (\mathcal{E}^u_3 - \kappa_3 t) + (\tilde{\mathcal{E}}^u_\psi(t) - \tilde{\kappa} t) + (\mathcal{E}^u_{1,1}(t) - \kappa_{1,1} t) + (\tilde{\mathcal{E}}^u_{1,\psi}(t) - \tilde{\kappa}_1 t).\end{aligned}$$

There exists a constant $\mathcal{C}$ depending on $a, \nu, h, \mathcal{B}_1$ such that

$$\begin{aligned}\mathcal{E}^u_\psi(t) - (\kappa_1 + \kappa_3 + \tilde{\kappa} + \kappa_{1,1} + \tilde{\kappa}_1)t - \mathcal{C}\left(1 + \|u_0\|^6 + \|w(1)\|^4\right) \\ \leq \left(\mathcal{E}^u_1 - \kappa_1 t - \mathcal{C}_1\|u_0\|^2\right) + \left(\mathcal{E}^u_3 - \kappa_3 t - \mathcal{C}_3\|u_0\|^6\right) \\ + \left(\mathcal{E}^u_{1,1} - \kappa_{1,1} t - \mathcal{C}_{1,1}\|w(1)\|^2\right) \\ + \left(\tilde{\mathcal{E}}^u_\psi(t) - \tilde{\kappa} t - \tilde{\mathcal{C}}\left(1 + \|u_0\|^6\right)\right) \\ + \left(\tilde{\mathcal{E}}^u_{1,\psi}(t) - \tilde{\kappa}_1 t - \tilde{\mathcal{C}}_1\left(1 + \|u_0\|^6 + \|w(1)\|^4\right)\right).\end{aligned}$$

Hence, setting

$$\kappa := \kappa_1 + \kappa_3 + \tilde{\kappa} + \kappa_{1,1} + \tilde{\kappa}_1, \qquad \gamma := \frac{\tilde{\gamma} \wedge \tilde{\gamma}_1}{5},$$

and applying Propositions 3.1, 3.2, 3.3, and 3.4, we conclude the proof. □

As a corollary, we now establish an estimate for the distribution function of the stopping time

$$\tau^u_1 := \inf\left\{t \geq 0 \,|\, \mathcal{E}^u_\psi(t) \geq (K + 2L)t + 2\rho + \mathscr{C}\left(1 + \|u_0\|^6\right)\right\}, \tag{3.24}$$

where $K, \mathscr{C}$, and L are some parameters to be determined later.

Corollary 3.6. *Let $\kappa, \gamma, \mathcal{C}$ be given in Proposition 3.5. If $K \geq \kappa$ and $\mathscr{C} \geq \mathcal{C}$, then*

$$\mathbb{P}\{l \leq \tau^u_1 < \infty\} \leq C_{a,\nu,h,\mathcal{B}_1,q}\left(\mathbb{E}\|u_0\|^{8(q+1)} + 1\right)\left(e^{-\gamma(\rho+Ll)} + \frac{1}{(\rho + Ll)^{\frac{q}{2}-1}}\right)$$

for any $q, \rho > 2$ and $L, l \geq 0$.

Proof. Note that

$$\begin{aligned}\{l \leq \tau^u_1 < \infty\} \subset \left(\{l \leq \tau^u_1 < \infty\} \bigcap \left\{\|w(1)\| \leq \mathcal{C}^{-\frac{1}{4}}(\rho + Ll)^{\frac{1}{4}}\right\}\right) \\ \bigcup \left\{\|w(1)\| > \mathcal{C}^{-\frac{1}{4}}(\rho + Ll)^{\frac{1}{4}}\right\}.\end{aligned}$$

By the definition of τ^u_1,

$$\begin{aligned}\mathcal{E}^u_\psi(\tau^u_1) &= (K + 2L)\,\tau^u_1 + 2\rho + \mathscr{C}\left(1 + \|u_0\|^6\right) \\ &\geq \kappa\tau^u_1 + \mathcal{C}\left(1 + \|u_0\|^6 + \|w(1)\|^4\right) + \rho + Ll\end{aligned}$$

on the event

$$\{l \leq \tau^u_1 < \infty\} \bigcap \left\{\|w(1)\| \leq \mathcal{C}^{-\frac{1}{4}}(\rho + Ll)^{\frac{1}{4}}\right\}.$$

Therefore,

$$\{l \le \tau_1^u < \infty\} \subset \left\{\|w(1)\| > \mathcal{C}^{-\frac{1}{4}}(\rho + Ll)^{\frac{1}{4}}\right\} \bigcup \left\{\sup_{t\ge 0}\left(\mathcal{E}_\psi^u(t) - \kappa t - \mathcal{C}\left(1 + \|u_0\|^6 + \|w(1)\|^4\right)\right) \ge \rho + Ll\right\}.$$

Applying Proposition 3.5 and Lemma A.1, we obtain the desired result. □

We introduce another stopping time τ_2^u in order to control the growth of $\mathcal{E}_1^u(t)$:

$$\tau_2^u := \inf\left\{t \ge 0 \,|\, \mathcal{E}_1^u(t) \ge (K+L)t + \rho + \mathscr{C}\|u_0\|^2\right\}. \tag{3.25}$$

Corollary 3.7. *Let $\kappa_1, \mathcal{C}_1$ be the constants in Proposition 3.1. If $K \ge \kappa_1$ and $\mathscr{C} \ge \mathcal{C}_1$, then*

$$\mathbb{P}\{l \le \tau_2^u < \infty\} \le C_{a,\nu,h,\mathcal{B}_0,q}\left(\mathbb{E}\|u_0\|^q + 1\right)\frac{1}{(\rho + Ll)^{\frac{q}{2}-1}}$$

for any $q, \rho > 2$ and $L, l \ge 0$.

Proof. Notice that

$$\mathcal{E}_1^u(\tau_2^u) = (K+L)\tau_2^u + \rho + \mathscr{C}\|u_0\|^2 \ge \kappa_1\tau_2^u + \mathcal{C}_1\|u_0\|^2 + \rho + Ll$$

on the event $\{l \le \tau_2^u < \infty\}$, which implies

$$\{l \le \tau_2^u < \infty\} \subset \left\{\sup_{t\ge 0}\left(\mathcal{E}_1^u(t) - \kappa_1 t - \mathcal{C}_1\|u_0\|^2\right) \ge \rho + Ll\right\}.$$

Hence, an application of Proposition 3.1 yields the result. □

Combining Corollaries 3.6 and 3.7, we can simultaneously control the growth of both $\mathcal{E}_\psi^u(t)$ and $\mathcal{E}_1^u(t)$ via the stopping time

$$\tau^u := \tau_1^u \wedge \tau_2^u. \tag{3.26}$$

Note that

$$\{l \le \tau^u < \infty\} \subset \{l \le \tau_1^u < \infty\} \bigcup \{l \le \tau_2^u < \infty\},$$

which allows to derive the following estimate for the distribution of τ^u.

Corollary 3.8. *Let $\kappa, \gamma, \mathcal{C}$ be given in Proposition 3.5 and $\kappa_1, \mathcal{C}_1$ in Proposition 3.1. If $K \ge \kappa \vee \kappa_1$ and $\mathscr{C} \ge \mathcal{C} \vee \mathcal{C}_1$, then*

$$\mathbb{P}\{l \le \tau^u < \infty\} \le C_{a,\nu,h,\mathcal{B}_1,q}\left(\mathbb{E}\|u_0\|^{8(q+1)} + 1\right)\left(e^{-\gamma(\rho+Ll)} + \frac{1}{(\rho + Ll)^{\frac{q}{2}-1}}\right)$$

for any $q, \rho > 2$ and $L, l \ge 0$.

4 Stability of solutions

We begin this section by establishing a Foiaş–Prodi type estimate for the NS system (1.1). We then derive a growth estimate for an auxiliary process that arises in the course of this estimate.

4.1 The Foiaş–Prodi type estimate

The Foiaş–Prodi type estimate is stated in Proposition 4.3. One of the key ingredients in its proof is the following truncated Poincaré inequality.

Lemma 4.1. *For any $\epsilon > 0$ and $A > 1$, there is an integer $N \geq 1$ such that*

$$\|\mathrm{Q}_N \chi_A f\| \leq \epsilon \|f\|_{H^{\frac{1}{2}}} \quad \text{for } f \in H^{\frac{1}{2}},$$

where $\chi_A : \mathbb{R}^2 \to [0,1]$ is a smooth cut-off function satisfying

$$\chi_A(x) = \begin{cases} 1, & x \in B(0,A), \\ 0, & x \notin B(0,2A) \end{cases}$$

with $B(0,A) := \{x \,|\, |x| < A\}$ and $\mathrm{Q}_N := \mathrm{I} - \mathrm{P}_N$ with P_N the same projector as in (2.8).

See Lemma 2.1 in [NZ24] for a proof of this lemma. We will also use the following weighted estimates, whose proof is deferred to the Appendix.

Lemma 4.2. *There is a constant $C > 0$ such that*

(a) *for any $f \in L^2(\mathbb{R}^2;\mathbb{R}^2)$,*

$$\|\psi(t)\Pi f\| \leq C\|\psi(t) f\|, \quad t \geq 2;$$

(b) *for any $u \in H^2$,*

$$\|\psi(t)\nabla^2 u\| \leq C\left(\|\nabla u\| + \|\psi(t)\nabla w\|\right), \quad t > 0,$$

where $w := \operatorname{curl} u$.

Proposition 4.3. *Let $g(t) := u(t) - v(t)$, where $\{u(t)\}_{t\geq 0}$ is the solution of (1.1) issued from $u \in H$ and $\{v(t)\}_{t\geq 0}$ is the solution of (2.8). Then, for any $\epsilon > 0$, there is a time $T \geq 2$ and an integer $N \geq 1$ such that*

$$\begin{aligned} \|g(t+T)\|^2 &\leq \left(\|g(s+T)\|^2 + C\|\mathrm{P}_N g(s+T)\|_{H^2}^2\right) e^{-a(t-s)} \\ &\times \exp\left(C\epsilon \int_{s+T}^{t+T} \left(\|v(r)\|_{H^1}^2 + \|\psi(r-1)\nabla v(r)\|^2 + \|\psi(r-1)v(r)\|^2\right) \mathrm{d}r\right) \\ &\times \exp\left(C\epsilon \int_{s+T}^{t+T} \left(\|w(r)\|_{H^1}^2 + \|\psi(r-1)\nabla u(r)\|^2 + \|\psi(r-1)\nabla w(r)\|^2\right) \mathrm{d}r\right) \end{aligned} \tag{4.1}$$

for any $t \geq s \geq 0$, where $C > 0$ is a constant depending on a, ν.

Proof. The equation for g is given by

$$\begin{cases} \partial_t g + ag + \mathrm{Q}_N[\Pi(u\cdot\nabla)u - \Pi(v\cdot\nabla)v - \nu\Delta g] = 0, \\ g|_{t=0} = u - u'. \end{cases} \tag{4.2}$$

Taking the inner product in H of this equation with g, we obtain

$$\begin{aligned} \frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}\|g(t)\|^2 + a\|g(t)\|^2 &= -\langle g, \mathrm{Q}_N[\Pi(g\cdot\nabla)u + \Pi(v\cdot\nabla)g - \nu\Delta g]\rangle \\ &=: -I_1 - I_2 - I_3. \end{aligned} \tag{4.3}$$

To estimate I_1, we decompose

$$I_1 = \langle g, \mathrm{Q}_N\chi_A\Pi(g\cdot\nabla)u\rangle + \langle \mathrm{Q}_N g, (1-\chi_A)\Pi(g\cdot\nabla)u\rangle =: I_{11} + I_{12}.$$

To bound I_{11}, we apply Lemma 4.1, the boundedness of the Leray projector in $H^{\frac{1}{2}}$, the Kato–Ponce inequality (e.g., see Proposition 1.1 in Chapter 2 of [Tay00])

$$\||\nabla|^{\frac{1}{2}}(f_1 f_2)\| \lesssim \||\nabla|^{\frac{1}{2}} f_1\|_{L^4}\|f_2\|_{L^4} + \||\nabla|^{\frac{1}{2}} f_2\|_{L^4}\|f_1\|_{L^4},$$

the embedding $H^{\frac{1}{2}} \hookrightarrow L^4$, and the estimate

$$\|\nabla^j u\| \lesssim \|\nabla^{j-1} w\|, \qquad \forall j \geq 1$$

to derive

$$\begin{aligned} I_{11} &\leq C\epsilon\|g\|\|(g\cdot\nabla)u\|_{H^{\frac{1}{2}}} \\ &\leq C\epsilon\|g\|\left(\|(g\cdot\nabla)u\| + \||\nabla|^{\frac{1}{2}} g\|_{L^4}\|\nabla u\|_{L^4} + \||\nabla|^{\frac{1}{2}}\nabla u\|_{L^4}\|g\|_{L^4}\right) \\ &\leq C\epsilon\|g\|\|g\|_{H^1}\|\nabla u\|_{H^1} \leq C\epsilon\|g\|\|g\|_{H^1}\|w\|_{H^1}. \end{aligned}$$

To bound I_{12}, we choose A and T sufficiently large such that

$$\frac{1}{|\psi(t-1,x)|} \leq \epsilon$$

for any $t \geq T$ and $x \notin B(0,A)$ and use properties (a) and (b) in Lemma 4.2:

$$\begin{aligned} I_{12} &\leq \epsilon\|g\|\|\tilde{\psi}\Pi(g\cdot\nabla)u\| \leq C\epsilon\|g\|\|\tilde{\psi}(g\cdot\nabla)u\| \\ &\leq C\epsilon\|g\|\|g\|_{H^1}\|\tilde{\psi}\nabla u\|_{H^1} \\ &\leq C\epsilon\|g\|\|g\|_{H^1}\left(\|w\|_{H^1} + \|\tilde{\psi}\nabla u\| + \|\tilde{\psi}\nabla w\|\right), \end{aligned}$$

where we denote $\tilde{\psi}(t) := \psi(t-1)$. Therefore,

$$I_1 \leq C\epsilon\|g\|\|g\|_{H^1}\left(\|w\|_{H^1} + \|\tilde{\psi}\nabla u\| + \|\tilde{\psi}\nabla w\|\right). \tag{4.4}$$

As for I_2, we use the cancellation property of the convection term and then decompose as in the case of I_1:

$$\begin{aligned} I_2 &= \langle g, \mathrm{Q}_N \Pi(v\cdot\nabla)\mathrm{P}_N g\rangle \\ &= \langle g, \mathrm{Q}_N \chi_A \Pi(v\cdot\nabla)\mathrm{P}_N g\rangle + \langle \mathrm{Q}_N g, (1-\chi_A)\Pi(v\cdot\nabla)\mathrm{P}_N g\rangle =: I_{21} + I_{22}. \end{aligned}$$

The estimates for I_{21} and I_{22} are similar to those for I_{11} and I_{12}:

$$I_{21} \le C\epsilon\|g\|\|(v\cdot\nabla)\mathrm{P}_N g\|_{H^{\frac{1}{2}}} \le C\epsilon\|g\|\|\nabla \mathrm{P}_N g\|_{H^1}\|v\|_{H^1}$$

and

$$\begin{aligned} I_{22} &\le \epsilon\|g\|\|\tilde{\psi}\Pi(v\cdot\nabla)\mathrm{P}_N g\| \le C\epsilon\|g\|\|\tilde{\psi}(v\cdot\nabla)\mathrm{P}_N g\| \\ &\le C\epsilon\|g\|\|\nabla \mathrm{P}_N g\|_{H^1}\left(\|\tilde{\psi}v\| + \|\tilde{\psi}\nabla v\| + \|v\|_{H^1}\right). \end{aligned}$$

Hence, we get

$$I_2 \le C\epsilon\|g\|\|\mathrm{P}_N g\|_{H^2}\left(\|\tilde{\psi}v\| + \|\tilde{\psi}\nabla v\| + \|v\|_{H^1}\right). \tag{4.5}$$

Finally, for I_3, we have

$$\begin{aligned} -I_3 &= \nu\langle g, \mathrm{Q}_N \Delta g\rangle = -\nu\|\nabla g\|^2 - \nu\langle g, \mathrm{P}_N \Delta g\rangle \\ &= -\nu\|\nabla g\|^2 - \nu\langle \Delta \mathrm{P}_N g, g\rangle \le -\nu\|\nabla g\|^2 + \nu\|\Delta \mathrm{P}_N g\|\|g\|. \end{aligned} \tag{4.6}$$

Plugging the estimates (4.4)–(4.6) into (4.3), we obtain

$$\begin{aligned} &\frac{\mathrm{d}}{\mathrm{d}t}\|g(t)\|^2 + a\|g\|^2 + \nu\|\nabla g\|^2 \lesssim_{a,\nu} \|\mathrm{P}_N g\|_{H^2}^2 \\ &\quad + \epsilon\|g\|^2\left(\|w\|_{H^1}^2 + \|\tilde{\psi}\nabla u\|^2 + \|\tilde{\psi}\nabla w\|^2 + \|\tilde{\psi}v\|^2 + \|\tilde{\psi}\nabla v\|^2 + \|v\|_{H^1}^2\right). \end{aligned}$$

An application of the Gronwall inequality, together with the fact that

$$\mathrm{P}_N g(t) = e^{-a(t-s)}\mathrm{P}_N g(s), \tag{4.7}$$

yields (4.1). □

4.2 Growth estimate for an auxiliary process

To handle the integral terms appearing in the Foiaş–Prodi type inequality (4.1), we establish an estimate for the stopping time τ^v defined in (3.26), where $\{v(t)\}$ is the solution of (2.8). In what follows, we assume that the constants K and $\mathscr{C}$ are sufficiently large so that Corollary 3.8 holds. The values of ρ and L will be specified later in Section 5.

Proposition 4.4. *There are constants $\gamma, C > 0$ and an integer $N \geq 1$ depending on $a, \nu, h, \mathcal{B}_1, \mathcal{B}_\varphi, L$ such that the following inequality holds*

$$\mathbb{P}\{\tau^v < \infty\} \leq C_{a,\nu,h,\mathcal{B}_1,q} \left(1 + d^{8(q+1)}\right) \left(e^{-\gamma\rho} + \frac{1}{\rho^{\frac{q}{2}-1}}\right) + \frac{1}{2}\left(\exp\left(Cd^2 e^{C(\rho+d^6)}\right) - 1\right)^{\frac{1}{2}} \tag{4.8}$$

for any $q, \rho > 2$, $d > 0$, and $u, u' \in B_H(0, d)$, provided that (1.7) holds. The constants γ, C and integer N do not depend on q, ρ, d, u, u'.

Proof. Step 1. We begin by reducing the proof to an estimate involving a truncated version of v. More precisely, we denote by $\{u(t)\}_{t\geq 0}$ the solution of the problem (1.1) starting from u, and consider the truncated process $\{\hat{u}(t)\}_{t\geq 0}$ defined as follows: for $t \leq \tau^u$, it coincides with $\{u(t)\}_{t\geq 0}$; and for $t \geq \tau^u$, it solves the linear equation

$$\partial_t z + az = \nu\Delta z. \tag{4.9}$$

We define the truncated processes $\{\hat{u}'(t)\}_{t\geq 0}$ and $\{\hat{v}(t)\}_{t\geq 0}$ in the same way with the stopping time τ^u replaced by $\tau^{u'}$ and τ^v, respectively. Then, we have

$$\{\tau^v < \infty\} = \{\tau^{\hat{v}} < \infty\}.$$

Thus, it remains to control the probability $\mathbb{P}\{\tau^{\hat{v}} < \infty\}$.
Step 2. We now reduce the bound on $\mathbb{P}\{\tau^{\hat{v}} < \infty\}$ to an estimate involving a certain measurable transformation. Without loss of generality, we assume that the probability space $(\Omega, \mathscr{F}, \mathbb{P})$ has the following structure: $\Omega := C_0([0,\infty); H)$ is the space of continuous functions $f : [0,\infty) \to H$ vanishing at $t = 0$, endowed with the topology of uniform convergence on compact sets, $\mathbb{P}$ is the distribution of the Wiener process

$$W(t) := \sum_{j=1}^{\infty} b_j \beta_j(t) e_j, \tag{4.10}$$

and $\mathscr{F}$ is the completion of the Borel σ-algebra of Ω. Now, let $N \geq 1$ be an integer (to be specified in Step 4) and define a transformation $\Phi^{u,u'} : \Omega \to \Omega$ as follows:

$$\omega_t \mapsto \omega_t - \int_0^t \mathbb{I}_{\{s\leq\tau^v\}} \mathrm{P}_N[\Pi(u\cdot\nabla)u - \Pi(v\cdot\nabla)v - \nu\Delta(u-v)]\mathrm{d}s. \tag{4.11}$$

The pathwise uniqueness for the system (1.1) implies that

$$\mathbb{P}\left\{\hat{u}'(t, \Phi^{u,u'}(\omega)) = \hat{v}(t,\omega) \text{ for any } t \geq 0\right\} = 1. \tag{4.12}$$

Thus,

$$\begin{aligned}\mathbb{P}\{\tau^{\hat{v}} < \infty\} = \Phi_*^{u,u'}\mathbb{P}\{\tau^{\hat{u}'} < \infty\} &\leq \mathbb{P}\{\tau^{\hat{u}'} < \infty\} + \|\mathbb{P} - \Phi_*^{u,u'}\mathbb{P}\|_{\mathrm{var}} \\ &= \mathbb{P}\{\tau^{u'} < \infty\} + \|\mathbb{P} - \Phi_*^{u,u'}\mathbb{P}\|_{\mathrm{var}},\end{aligned} \tag{4.13}$$

where $\Phi_*^{u,u'}\mathbb{P}$ is the push-forward of $\mathbb{P}$ under $\Phi^{u,u'}$. The probability $\mathbb{P}\{\tau^{u'} < \infty\}$ is estimated in Corollary 3.8.

Step 3. To bound the term $\|\mathbb{P} - \Phi_*^{u,u'}\mathbb{P}\|_{\text{var}}$ in (4.13), we employ the strategy in Section 3.3.3 of [KS12] based on the Girsanov theorem. Define an auxiliary transformation $\tilde{\Phi}^{u,u'} : \Omega \to \Omega$ by

$$\omega_t \mapsto \omega_t - \int_0^t \mathbb{I}_{\{s \le \tau^v \wedge \tau^u\}} \mathrm{P}_N[\Pi(u \cdot \nabla)u - \Pi(v \cdot \nabla)v - \nu\Delta(u - v)]\mathrm{d}s. \tag{4.14}$$

By the definition of total variation metric, it follows that

$$\begin{aligned}\|\mathbb{P} - \Phi_*^{u,u'}\mathbb{P}\|_{\text{var}} &\le \|\mathbb{P} - \tilde{\Phi}_*^{u,u'}\mathbb{P}\|_{\text{var}} + \|\tilde{\Phi}_*^{u,u'}\mathbb{P} - \Phi_*^{u,u'}\mathbb{P}\|_{\text{var}} \\ &\le \|\mathbb{P} - \tilde{\Phi}_*^{u,u'}\mathbb{P}\|_{\text{var}} + \mathbb{P}\{\tilde{\Phi}^{u,u'} \neq \Phi^{u,u'}\} \\ &\le \|\mathbb{P} - \tilde{\Phi}_*^{u,u'}\mathbb{P}\|_{\text{var}} + \mathbb{P}\{\tau^u < \infty\}. \end{aligned} \tag{4.15}$$

Therefore, it remains to estimate $\|\mathbb{P} - \tilde{\Phi}_*^{u,u'}\mathbb{P}\|_{\text{var}}$. To this end, we introduce the following decomposition of Ω:

$$\Omega = C([0,\infty); \mathrm{P}_N H) \oplus C([0,\infty); \mathrm{Q}_N H),$$

so that any $\omega \in \Omega$ can be expressed as $\omega = (\omega^{(1)}, \omega^{(2)})$. Accordingly, the transformation $\tilde{\Phi}^{u,u'}$ in (4.14) can be represented in the form

$$\tilde{\Phi}^{u,u'}(\omega^{(1)}, \omega^{(2)}) = (\Psi^{u,u'}(\omega^{(1)}, \omega^{(2)}), \omega^{(2)}),$$

with $\Psi^{u,u'} : \Omega \to C([0,\infty); \mathrm{P}_N H)$ given by

$$\Psi^{u,u'}(\omega^{(1)}, \omega^{(2)})_t := \omega_t^{(1)} + \int_0^t \mathcal{A}(s; \omega^{(1)}, \omega^{(2)})\mathrm{d}s,$$

where

$$\mathcal{A}(t) := -\mathbb{I}_{\{t \le \tau^v \wedge \tau^u\}} \mathrm{P}_N[\Pi(u \cdot \nabla)u - \Pi(v \cdot \nabla)v - \nu\Delta(u - v)]. \tag{4.16}$$

Let $\mathbb{P}_N := (\mathrm{P}_N)_*\mathbb{P}$ and $\mathbb{P}_N^\perp := (\mathrm{Q}_N)_*\mathbb{P}$, with P_N and Q_N being now the projections

$$\begin{aligned}\mathrm{P}_N : \Omega :&\to C_0([0,\infty); \mathrm{P}_N H), \\ \mathrm{Q}_N : \Omega :&\to C_0([0,\infty); \mathrm{Q}_N H).\end{aligned}$$

Lemma 3.3.13 in [KS12] implies that

$$\|\mathbb{P} - \tilde{\Phi}_*^{u,u'}\mathbb{P}\|_{\text{var}} \le \int_{C_0([0,\infty);\mathrm{Q}_N H)} \|\Psi_*^{u,u'}(\mathbb{P}_N, \omega^{(2)}) - \mathbb{P}_N\|_{\text{var}} \mathbb{P}_N^\perp(\mathrm{d}\omega^{(2)}). \tag{4.17}$$

By the Girsanov theorem, for each $\omega^{(2)}$, we have

$$\begin{aligned}&\|\Psi_*^{u,u'}(\mathbb{P}_N, \omega^{(2)}) - \mathbb{P}_N\|_{\text{var}} \\ &\qquad \le \frac{1}{2}\left(\left(\mathbb{E}_N \exp\left(6 \sup_{1 \le j \le N} b_j^{-2} \int_0^\infty \|\mathcal{A}(t; \cdot, \omega^{(2)})\|^2 \mathrm{d}t\right)\right)^{\frac{1}{2}} - 1\right)^{\frac{1}{2}},\end{aligned} \tag{4.18}$$

provided that the Novikov condition

$$\mathbb{E}_N \exp\left(c\int_0^\infty \|\mathcal{A}(t;\cdot,\omega^{(2)})\|^2 \mathrm{d}t\right) < \infty \tag{4.19}$$

is satisfied for any $c>0$ and $\omega^{(2)}$, where $\mathbb{E}_N$ is expectation with respect to $\mathbb{P}_N$. *Step 4.* To verify that the Novikov condition holds, we need to establish a pathwise estimate for $\|g(t)\|^2$ up to the stopping time $\tau^u \wedge \tau^v$, where

$$g(t) := u(t) - v(t) = \hat{u}(t) - \hat{v}(t), \quad t \in [0, \tau^u \wedge \tau^v).$$

Let C_1 be the constant in (4.1). By Proposition 4.3 applied for

$$\epsilon := \frac{a}{4C_1(K+2L)}, \tag{4.20}$$

there are $T \geq 2$ and $N \in \mathbb{N}_+$ such that (4.1) holds. We consider two cases. *Case 1:* $\tau^u \wedge \tau^v \leq T$. Using the equality (4.3), the estimate (4.6), the nonlinear estimates

$$\begin{aligned}
\langle \mathrm{Q}_N g, \Pi(g\cdot\nabla)u\rangle &= \langle \mathrm{Q}_N g, (g\cdot\nabla)u\rangle \\
&\leq \|\nabla u\|\|g\|_{L^4}\|\mathrm{Q}_N g\|_{L^4} \\
&\leq \|\nabla u\|\|g\|_{L^4}^2 + \|\nabla u\|\|g\|_{L^4}\|\mathrm{P}_N g\|_{L^4} \\
&\leq C\|\nabla u\|\|g\|\|\nabla g\| + C\|\nabla u\|\|g\|\|\nabla \mathrm{P}_N g\|^{\frac{1}{2}}\|\nabla g\|^{\frac{1}{2}} \\
&\leq \frac{\nu}{8}\|\nabla g\|^2 + C_\nu\|g\|^2\|\nabla u\|^2 + C_\nu\|\nabla \mathrm{P}_N g\|^2
\end{aligned}$$

and

$$\begin{aligned}
\langle \mathrm{Q}_N g, \Pi(v\cdot\nabla)g\rangle &= \langle \mathrm{Q}_N g, (v\cdot\nabla)\mathrm{P}_N g\rangle \\
&\leq C\|g\|\|v\|_{H^1}\|\mathrm{P}_N g\|_{H^2} \leq C\|g\|^2\|v\|_{H^1}^2 + C\|\mathrm{P}_N g\|_{H^2}^2,
\end{aligned}$$

the Gronwall inequality, and (4.7), we get

$$\begin{aligned}
\|g(t)\|^2 \leq{}& \left(\|u-u'\|^2 + C_{a,\nu}\|\mathrm{P}_N(u-u')\|_{H^2}^2\right) \\
&\times \exp\left(-at + C_{a,\nu}\int_0^t \left(\|u\|_{H^1}^2 + \|v\|_{H^1}^2\right)\mathrm{d}s\right).
\end{aligned} \tag{4.21}$$

By the definition of $\tau^u \wedge \tau^v$,

$$\mathcal{E}_\psi^u(t) < (K+2L)t + 2\rho + \mathscr{C}(1+\|u\|^6) \leq (K+2L)t + 2\rho + \mathscr{C}(1+d^6), \tag{4.22}$$

$$\mathcal{E}_\psi^v(t) < (K+2L)t + 2\rho + \mathscr{C}(1+\|u'\|^6) \leq (K+2L)t + 2\rho + \mathscr{C}(1+d^6) \tag{4.23}$$

for $t < \tau^u \wedge \tau^v$. Hence,

$$\begin{aligned}
\|g(t)\|^2 \leq{}& \left(\|u-u'\|^2 + C_{a,\nu}\|\mathrm{P}_N(u-u')\|_{H^2}^2\right) \\
&\times \exp\left(-at + C_{a,\nu}[(K+2L)T + 2\rho + \mathscr{C}(1+d^6)]\right).
\end{aligned} \tag{4.24}$$

As

$$\|\mathrm{P}_N(u-u')\|_{H^2}^2 \lesssim_N \sum_{j=1}^N |\langle u-u', e_j\rangle|^2 \|e_j\|_{H^2}^2 \lesssim_N \|u-u'\|^2,$$

it follows from (4.24) that

$$\|g(t)\|^2 \le Cd^2 \exp(-at + C(\rho + d^6)), \tag{4.25}$$

where C is a constant depending on $a, \nu, h, \mathcal{B}_1, \mathcal{B}_\varphi, L$.
Case 2: $\tau^u \wedge \tau^v > T$. Inequality (4.25) holds true for $t \in [0,T]$. To estimate $\|g(t)\|^2$ over $[T, \tau^u \wedge \tau^v)$, we apply Proposition 4.3 with same ϵ as in (4.20) and use (4.7), (4.22), and (4.23):

$$\begin{aligned}\|g(t)\|^2 &\le \left(\|g(T)\|^2 + C_1\|\mathrm{P}_N g(T)\|_{H^2}^2\right) \exp\left(-a(t-T) + C_1\epsilon(\mathcal{E}_\psi^u(t) + \mathcal{E}_\psi^v(t))\right) \\ &\le \left(\|g(T)\|^2 + C_1\|\mathrm{P}_N g(T)\|_{H^2}^2\right) \exp\left(-\frac{a}{2}t + aT + \frac{a(\rho + \mathscr{C}(1+d^6))}{(K+2L)}\right)\end{aligned}$$

for $t \in [T, \tau^u \wedge \tau^v)$. Therefore, we obtain the bound

$$\|g(t)\|^2 \le Cd^2 \exp\left(-\frac{a}{2}t + C(\rho + d^6)\right), \quad t \in [T, \tau^u \wedge \tau^v). \tag{4.26}$$

Step 5. To verify the Novikov condition (4.19), we first bound the terms on the right-hand side of (4.16). First, integrating by parts, we get

$$\|\mathrm{P}_N \Delta(u-v)\|^2 = \sum_{j=1}^N \langle \Delta g, e_j\rangle^2 = \sum_{j=1}^N \langle g, \Delta e_j\rangle^2 \lesssim_N \|g\|^2. \tag{4.27}$$

To estimate the term $\mathrm{P}_N[\Pi(u\cdot\nabla)u - \Pi(v\cdot\nabla)v]$, we use the identity

$$(u\cdot\nabla)u = \mathrm{div}(u\otimes u),$$

where $\otimes$ denotes the tensor product of vector fields:

$$\begin{aligned}\|\mathrm{P}_N[\Pi(u\cdot\nabla)u - \Pi(v\cdot\nabla)v]\|^2 &= \sum_{j=1}^N \langle \mathrm{div}\left(u\otimes u - v\otimes v\right), e_j\rangle^2 \\ &\lesssim \sum_{j=1}^N \left(\langle g\otimes u, \nabla e_j\rangle^2 + \langle v\otimes g, \nabla e_j\rangle^2\right) \\ &\lesssim \sum_{j=1}^N \left(\|g\|^2\|u\|_{H^1}^2\|\nabla e_j\|_{H^1}^2 + \|g\|^2\|v\|_{H^1}^2\|\nabla e_j\|_{H^1}^2\right) \\ &\lesssim_N \|g\|^2\left(\|u\|_{H^1}^2 + \|v\|_{H^1}^2\right). \end{aligned} \tag{4.28}$$

Combining (4.27) and (4.28), we derive

$$\|\mathcal{A}(t)\|^2 \lesssim_N \mathbb{I}_{\{t\le\tau^u\wedge\tau^v\}}\|g\|^2\left(1 + \|u\|_{H^1}^2 + \|v\|_{H^1}^2\right). \tag{4.29}$$

To further estimate the terms on the right-hand side of this inequality, we use (4.26):

$$\int_0^{\tau^u\wedge\tau^v} \|g(t)\|^2 \mathrm{d}t \le Cd^2 e^{C\left(\rho+d^6\right)}. \tag{4.30}$$

Furthermore, integrating by parts, one has

$$\begin{aligned}
&\int_0^{\tau^u\wedge\tau^v} \|g(t)\|^2 \left(\|u\|_{H^1}^2 + \|v\|_{H^1}^2\right) \mathrm{d}t \\
&\qquad \le Cd^2 e^{C(\rho+d^6)} \int_0^{\tau^u\wedge\tau^v} e^{-\frac{at}{2}} \mathrm{d} \int_0^t \left(\|u\|_{H^1}^2 + \|v\|_{H^1}^2\right) \mathrm{d}s \\
&\qquad \le Cd^2 e^{C(\rho+d^6)} e^{-\frac{a\tau^u\wedge\tau^v}{2}} \left(\mathcal{E}_\psi^u(\tau^u\wedge\tau^v) + \mathcal{E}_\psi^v(\tau^u\wedge\tau^v)\right) \\
&\qquad\quad + Cd^2 e^{C(\rho+d^6)} \frac{a}{2} \int_0^{\tau^u\wedge\tau^v} e^{-\frac{at}{2}} \left(\mathcal{E}_\psi^u(t) + \mathcal{E}_\psi^v(t)\right) \mathrm{d}t \\
&\qquad \le Cd^2 e^{C(\rho+d^6)}.
\end{aligned} \tag{4.31}$$

Finally, (4.30) and (4.31) imply the desired Novikov condition:

$$\mathbb{E}_N \exp\left(c\int_0^\infty \|\mathcal{A}(t;\cdot,\omega^{(2)})\|^2 \mathrm{d}t\right) \le \exp\left(cCd^2 e^{C\left(\rho+d^6\right)}\right). \tag{4.32}$$

Besides, combining the estimates (4.17), (4.18), and (4.32), we get

$$\|\mathbb{P} - \tilde{\Phi}_*^{u,u'}\mathbb{P}\|_{\mathrm{var}} \le \frac{1}{2}\left(\exp\left(Cd^2 e^{C\left(\rho+d^6\right)}\right) - 1\right)^{\frac{1}{2}}. \tag{4.33}$$

This, together with (4.13), (4.15), and Corollary 3.8, implies (4.8). □

5 Proof of Theorem 2.2

Here we establish Theorem 2.2, which, as previously mentioned, implies Theorem 2.1 and, consequently, the Main Theorem.

5.1 Recurrence

Before verifying the recurrence property, we first demonstrate that the Markov family $(u(t), \mathbb{P}_u)$ associated with (1.1) is irreducible. Recall that $P_t(u,\Gamma) := \mathbb{P}\{S_t(u,\cdot)\in\Gamma\}$ is the transition function of $(u(t),\mathbb{P}_u)$.

Lemma 5.1. *Let $N \ge 1$ be an arbitrary fixed integer. For any $R, d > 0$, there exist constants $p, T > 0$ depending on $R, d, a, \nu, h, \mathcal{B}_1$ such that*

$$P_T(u_0, B_H(0,d)) \ge p \tag{5.1}$$

for all $u_0 \in B_H(0,R)$, provided that (1.7) holds and h belongs to the space spanned by the family $\{e_1, e_2, \dots, e_N\}$.

Proof. Let us define

$$u^1(t) := u(t) - W_h(t),$$

where $W_h(t) := th + W(t)$ with $W(t)$ being the Wiener process given in (4.10). Then, u^1 solves the equation

$$\partial_t u^1 + \Pi(u \cdot \nabla)u - \nu\Delta u + a(u^1 + W_h) = 0.$$

Taking the inner product in H of this equation with u^1, we get

$$\begin{aligned}\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}\|u^1\|^2 + a\|u^1\|^2 &= \langle -\Pi(u \cdot \nabla)u + \nu\Delta u, u^1\rangle - a\langle W_h, u^1\rangle \\ &=: I_4 + I_5 + I_6. \end{aligned} \tag{5.2}$$

By the cancellation property of the convection term, it follows that

$$I_4 = -\langle \Pi(u \cdot \nabla)u, u^1\rangle = \langle (u \cdot \nabla)u, W_h\rangle \le \|u\|_{H^1}^2 \|W_h\|_{H^1} \tag{5.3}$$

and

$$\begin{aligned} I_5 &= -\nu\langle \nabla u, \nabla u^1\rangle = -\nu\|\nabla u^1\|^2 - \nu\langle \nabla W_h, \nabla u^1\rangle \\ &\le -\nu\|\nabla u^1\|^2 + \nu\|W_h\|_{H^1}\|\nabla u^1\| \\ &\le -\frac{\nu}{2}\|\nabla u^1\|^2 + C_\nu\|W_h\|_{H^1}^2 \end{aligned} \tag{5.4}$$

and

$$I_6 \le a\|W_h\|\|u^1\| \le \frac{a}{2}\|u^1\|^2 + C_a\|W_h\|^2. \tag{5.5}$$

Combining (5.2)–(5.5), we obtain

$$\begin{aligned}&\frac{\mathrm{d}}{\mathrm{d}t}\|u^1\|^2 + a\|u^1\|^2 + \nu\|\nabla u^1\|^2 \\ &\qquad \le C_{a,\nu}\left(\|u^1\|_{H^1}^2\|W_h\|_{H^1} + \|W_h\|_{H^1}^3 + \|W_h\|_{H^1}^2\right). \end{aligned} \tag{5.6}$$

Therefore,

$$\frac{\mathrm{d}}{\mathrm{d}t}\|u^1\|^2 + \frac{a}{2}\|u^1\|^2 \le \epsilon \tag{5.7}$$

on the event

$$\Omega_{T,\epsilon} := \left\{ \sup_{t\in[0,T]} \|W_h\|_{H^1} \le \min\left\{ \frac{a \wedge \nu}{2C_{a,\nu}}, \sqrt{\frac{\epsilon}{2C_{a,\nu}}}, \left(\frac{\epsilon}{2C_{a,\nu}}\right)^{\frac{1}{3}} \right\} \right\}.$$

This implies that

$$\|u^1(T)\|^2 \le C_a\epsilon + e^{-\frac{aT}{2}}\|u_0\|^2. \tag{5.8}$$

Choosing T large and ϵ small enough, we get

$$u^1(T), W_h(T) \in B_H(0, d/2)$$

on $\Omega_{T,\epsilon}$, so $u(T) \in B_H(0, d)$ for any $u_0 \in B_H(0, R)$. Notice that $p := \mathbb{P}(\Omega_{T,\epsilon}) > 0$, provided that (1.7) holds and h belongs to the space spanned by the family $\{e_1, e_2, \dots, e_N\}$. This leads to the desired estimate (5.1). □

Remark 5.2. The assumption that h belongs to the space spanned by the family $\{e_1, e_2, \dots, e_N\}$ is required to have the positivity of $\mathbb{P}(\Omega_{T,\epsilon})$. It is worth noting that this positivity can also be achieved under the conditions in (1.8); see Lemma 3.1 in [NZ24] for a proof. However, in this case, the number N of non-vanishing modes will depend on the convergence rate q, as the integer N depends on the parameter d, which itself depends on q; see the proof of Lemma 5.5 for further details.

Combining Lemma 5.1 with Propositions 3.5 and 4.3 and using the coupling construction of the extension $(\boldsymbol{u}(t), \mathbb{P}_{\boldsymbol{u}})$, one can deduce that this extension is also irreducible. As the argument mirrors the proof for the complex Ginzburg–Landau equation case (cf. Proposition 4.2 in [NZ24]), we omit the details.

Lemma 5.3. *There is an integer $N \geq 1$ depending on $a, \nu, h, \mathcal{B}_1, \mathcal{B}_\varphi$ such that for any $R, d > 0$, there are constants $p, T > 0$ depending on $R, d, a, \nu, h, \mathcal{B}_1, \mathcal{B}_\varphi$ and satisfying*

$$\mathbb{P}_{\boldsymbol{u}}\{\boldsymbol{u}(T) \in B_H(0,d) \times B_H(0,d)\} \geq p \tag{5.9}$$

for all $\boldsymbol{u} \in B_H(0,R) \times B_H(0,R)$, provided that (1.7) holds and h belongs to the space spanned by the family $\{e_1, e_2, \dots, e_N\}$.

From the estimate (A.1), it follows that $F(u) := 1 + \|u\|^2$ is a Lyapunov function for the family $(u(t), \mathbb{P}_u)$. Combining this and the irreducibility property in the previous proposition, we obtain the recurrence property (2.8), as explained in Section 3 in [Shi08]. To summarize, we have the following result.

Proposition 5.4. *There exists an integer $N \geq 1$ depending on the parameters $a, \nu, h, \mathcal{B}_1, \mathcal{B}_\varphi$ such that for any $d > 0$, there are constants $\delta, T, C > 0$ depending on $d, a, \nu, h, \mathcal{B}_1, \mathcal{B}_\varphi$ and satisfying*

$$\mathbb{E}_{\boldsymbol{u}} \exp(\delta \tau_d) \leq C\left(1 + \|u\|^2 + \|u'\|^2\right) \quad \text{for } \boldsymbol{u} \in H \times H, \tag{5.10}$$

provided that (1.7) holds and h belongs to the space spanned by the family $\{e_1, e_2, \dots, e_N\}$. Here, the stopping time τ_d is defined in (2.3).

5.2 Polynomial squeezing

In this subsection, we show that the polynomial squeezing property is verified for the extension $(\boldsymbol{u}(t), \mathbb{P}_{\boldsymbol{u}})$. We continue to denote by $u, u', v, \tilde{u}, \tilde{u}', \tilde{v}$ the processes constructed in Section 2 and by $\hat{u}, \hat{u}', \hat{v}$ the truncated processes appearing in the proof of Proposition 4.4. The constants $K, \gamma, \mathscr{C}$ are such that Corollary 3.8 holds.

The stopping time σ in Theorem 2.2 is defined by $\sigma := \tilde{\tau} \wedge \sigma_1$, where $\tilde{\tau} := \tau^{\tilde{u}} \wedge \tau^{\tilde{u}'}$ and

$$\sigma_1 := \inf\{t \geq 0 \,|\, \tilde{u}'(t) \neq \tilde{v}(t)\}.$$

Let us take any $T > 0$ and consider the events

$$\begin{aligned} \mathcal{Q}'_k &:= \{kT \leq \sigma < (k+1)T,\, \sigma_1 \geq \tilde{\tau}\}, \\ \mathcal{Q}''_k &:= \{kT \leq \sigma < (k+1)T,\, \sigma_1 < \tilde{\tau}\}, \quad k \geq 0. \end{aligned}$$

Before verifying the polynomial squeezing property for σ, we prove the following lemma.

Lemma 5.5. *There exists an integer $N \geq 1$ depending on the parameters $a, \nu, h, \mathcal{B}_1, \mathcal{B}_\varphi$ and a universal constant $L > 0$ such that for any $q > 1$, there are constants $d, T, \rho > 0$ depending on $q, a, \nu, h, \mathcal{B}_1, \mathcal{B}_\varphi$ and satisfying*

$$\mathbb{P}_{\boldsymbol{u}}(\mathcal{Q}_k') \vee \mathbb{P}_{\boldsymbol{u}}(\mathcal{Q}_k^{''}) \leq \frac{q-1}{2q}(k+1)^{-4q}, \quad k \geq 0$$

for any $\boldsymbol{u} \in \overline{B}_H(0,d) \times \overline{B}_H(0,d)$, provided that (1.7) holds.

Proof. Step 1: estimate for $\mathbb{P}_{\boldsymbol{u}}(\mathcal{Q}_k')$. Applying Corollary 3.8 and taking $d \leq 1$, we have

$$\mathbb{P}_{\boldsymbol{u}}(\mathcal{Q}_k') \leq \mathbb{P}_{\boldsymbol{u}}\{kT \leq \tilde{\tau} < \infty\} \lesssim_{q,a,\nu,h,\mathcal{B}_1,\mathcal{B}_\varphi} \frac{1}{(\rho + LkT)^{4q}},$$

where by choosing ρ, T sufficiently large and $L = 1$, we obtain

$$\mathbb{P}_{\boldsymbol{u}}(\mathcal{Q}_k') \leq \frac{q-1}{2q}(k+1)^{-4q}, \quad k \geq 0. \tag{5.11}$$

Step 2: estimate for $\mathbb{P}_{\boldsymbol{u}}(\mathcal{Q}_0^{''})$. Using the fact that $\{\tilde{v}(t)\}_{t\in[0,T]}$ and $\{\tilde{u}'(t)\}_{t\in[0,T]}$ are flows of the maximal coupling $(\mathcal{V}_T(u,u'), \mathcal{V}_T'(u,u'))$, we get

$$\begin{aligned}\mathbb{P}_{\boldsymbol{u}}(\mathcal{Q}_0^{''}) &\leq \mathbb{P}_{\boldsymbol{u}}\{0 \leq \sigma_1 \leq T\} = \mathbb{P}_{\boldsymbol{u}}\{\tilde{u}'(t) \neq \tilde{v}(t) \text{ for some } t \in [0,T]\} \\ &= \|\lambda_T(u,u') - \lambda_T'(u,u')\|_{\mathrm{var}},\end{aligned} \tag{5.12}$$

where $\lambda_T(u,u')$ and $\lambda_T'(u,u')$ are the distributions of $\{v(t)\}_{t\in[0,T]}$, $\{u'(t)\}_{t\in[0,T]}$. The term on the right-hand side of (5.12) is estimated as follows

$$\begin{aligned}&\|\lambda_T(u,u') - \lambda_T'(u,u')\|_{\mathrm{var}} \\ &\quad= \sup_{\Gamma\in\mathscr{B}(C([0,T];H))} |\mathbb{P}\{v(\cdot)\in\Gamma\} - \mathbb{P}\{u'(\cdot)\in\Gamma\}| \\ &\quad\leq \mathbb{P}\{\tau^v < \infty\} + \mathbb{P}\{\tau^{u'} < \infty\} \\ &\qquad+ \sup_{\Gamma\in\mathscr{B}(C([0,T];H))} |\mathbb{P}\{v(\cdot)\in\Gamma, \tau^v = \infty\} - \mathbb{P}\{u'(\cdot)\in\Gamma, \tau^{u'} = \infty\}| \\ &\quad=: \mathcal{P}_1 + \mathcal{P}_2 + \mathcal{P}_3,\end{aligned} \tag{5.13}$$

where the term $\mathcal{P}_3$ is estimated by

$$\begin{aligned}\mathcal{P}_3 &= \sup_{\Gamma\in\mathscr{B}(C([0,T];H))} |\mathbb{P}\{\hat{v}(\cdot)\in\Gamma, \tau^v = \infty\} - \mathbb{P}\{\hat{u}'(\cdot)\in\Gamma, \tau^{u'} = \infty\}| \\ &\leq \sup_{\Gamma\in\mathscr{B}(C([0,T];H))} |\mathbb{P}\{\hat{v}(\cdot)\in\Gamma\} - \mathbb{P}\{\hat{u}'(\cdot)\in\Gamma\}| \leq \|\mathbb{P} - \Phi_*^{u,u'}\mathbb{P}\|_{\mathrm{var}}.\end{aligned} \tag{5.14}$$

Here, we used the definitions of $\hat{u}'$, $\hat{v}$ and the equality (4.12), and $\Phi^{u,u'}$ is the transformation given in (4.11). Let $\tilde{\Phi}^{u,u'}$ be the auxiliary transformation defined in (4.14). Then, by combining (5.12)–(5.14) with (4.15), one has

$$\mathbb{P}_{\boldsymbol{u}}(\mathcal{Q}_0^{''}) \leq \|\mathbb{P} - \tilde{\Phi}_*^{u,u'}\mathbb{P}\|_{\mathrm{var}} + \mathbb{P}\{\tau^u < \infty\} + \mathbb{P}\{\tau^{u'} < \infty\} + \mathbb{P}\{\tau^v < \infty\},$$

which, together with (4.33), Proposition 4.4, and Corollary 3.8, implies

$$\mathbb{P}_{\boldsymbol{u}}(\mathcal{Q}_0^{''}) \leq \frac{q-1}{4q} + \left(\exp\left(Cd^2 e^{C\left(\rho+d^6\right)}\right) - 1\right)^{\frac{1}{2}}, \tag{5.15}$$

where C is a constant depending on $a, \nu, h, \mathcal{B}_1, \mathcal{B}_\varphi$. Choosing $d \leq 1$ sufficiently small such that

$$\left(\exp\left(Cd^2 e^{C\left(\rho+d^6\right)}\right) - 1\right)^{\frac{1}{2}} \leq \frac{q-1}{4q},$$

we conclude that

$$\mathbb{P}_{\boldsymbol{u}}(\mathcal{Q}_0^{''}) \leq \frac{q-1}{2q}$$

as desired.

Step 3: estimate for $\mathbb{P}_{\boldsymbol{u}}(\mathcal{Q}_k^{''})$, $k \geq 1$. In the case $k \geq 1$, the estimate is proved by using the Markov property and the arguments used in Step 2. We begin by applying the Markov property:

$$\begin{aligned}\mathbb{P}_{\boldsymbol{u}}(\mathcal{Q}_k^{''}) = \mathbb{P}_{\boldsymbol{u}}(\mathcal{Q}_k^{''}, \sigma \geq kT) &= \mathbb{E}_{\boldsymbol{u}}\left(\mathbb{I}_{\{\sigma \geq kT\}} \mathbb{E}_{\boldsymbol{u}}\left(\mathbb{I}_{\mathcal{Q}_k^{''}} \mid \mathscr{F}_{kT}\right)\right) \\ &\leq \mathbb{E}_{\boldsymbol{u}}\left(\mathbb{I}_{\{\sigma \geq kT\}} \mathbb{P}_{\boldsymbol{u}(kT)}\{0 \leq \sigma_1 \leq T\}\right).\end{aligned}$$

where, by proceeding similarly as in Step 2, we have

$$\begin{aligned}\mathbb{P}_{\boldsymbol{u}(kT)}\{0 \leq \sigma_1 \leq T\} \leq{}& \mathbb{P}_{\boldsymbol{u}(kT)}\{\tau^u < \infty\} + \mathbb{P}_{\boldsymbol{u}(kT)}\{\tau^{u'} < \infty\} \\ &+ \mathbb{P}_{\boldsymbol{u}(kT)}\{\tau^v < \infty\} + \|\mathbb{P} - \tilde{\Phi}_*^{\tilde{u}(kT), \tilde{u}'(kT)}\mathbb{P}\|_{\mathrm{var}}.\end{aligned}$$

Here, with a slight abuse of notation, we denote by $\tau^u, \tau^{u'}, \tau^v$ the stopping times that are defined in (3.26), but with the parameter ρ replaced by $\rho + \epsilon' akT$ for some ϵ' to be specified later. Combining this with (4.13) and (4.15), we see that

$$\begin{aligned}\mathbb{P}_{\boldsymbol{u}(kT)}\{0 \leq \sigma_1 \leq T\} \leq{}& 2\mathbb{P}_{\boldsymbol{u}(kT)}\{\tau^u < \infty\} + 2\mathbb{P}_{\boldsymbol{u}(kT)}\{\tau^{u'} < \infty\} \\ &+ 2\|\mathbb{P} - \tilde{\Phi}_*^{\tilde{u}(kT), \tilde{u}'(kT)}\mathbb{P}\|_{\mathrm{var}}.\end{aligned}$$

Therefore, it follows

$$\begin{aligned}\mathbb{P}_{\boldsymbol{u}}(\mathcal{Q}_k^{''}) \leq{}& 2\mathbb{E}_{\boldsymbol{u}}\left(\mathbb{I}_{\{\sigma \geq kT\}} \|\mathbb{P} - \tilde{\Phi}_*^{\tilde{u}(kT), \tilde{u}'(kT)}\mathbb{P}\|_{\mathrm{var}}\right) \\ &+ 2\mathbb{E}_{\boldsymbol{u}}\left(\mathbb{I}_{\{\sigma \geq kT\}} \mathbb{P}_{\boldsymbol{u}(kT)}\{\tau^u < \infty\}\right) \\ &+ 2\mathbb{E}_{\boldsymbol{u}}\left(\mathbb{I}_{\{\sigma \geq kT\}} \mathbb{P}_{\boldsymbol{u}(kT)}\{\tau^{u'} < \infty\}\right) \\ =:{}& 2I^1 + 2I^2 + 2I^3.\end{aligned} \tag{5.16}$$

To bound the term I^1, we proceed as in the proof of Proposition 4.4. Note that

$$\begin{aligned}&\tilde{\Phi}^{\tilde{u}(kT), \tilde{u}'(kT)}(\omega)_t \\ &\quad = \omega_t - \int_0^t \mathbb{I}_{\{s \leq \tau^k\}} \mathrm{P}_N[\Pi(u_k \cdot \nabla)u_k - \Pi(v_k \cdot \nabla)v_k - \nu\Delta(u_k - v_k)]\mathrm{d}s,\end{aligned}$$

where u_k, u_k', v_k are the solutions of (1.1) and (2.8) issued from $\tilde{u}(kT), \tilde{u}'(kT)$, and $\tilde{u}'(kT)$, and $\tau^k := \tau^{u_k} \wedge \tau^{v_k}$. In view of (4.17) and (4.18), we need to study the term $\int_0^\infty \|\mathcal{A}_k(t)\|^2 \mathrm{d}t$, where

$$\mathcal{A}_k(t) := -\mathbb{I}_{\{t \le \tau^k\}} \mathrm{P}_N[\Pi(u_k \cdot \nabla)u_k - \Pi(v_k \cdot \nabla)v_k - \nu\Delta(u_k - v_k)].$$

As before, we need to get a pathwise estimate for $g_k(t) := u_k(t) - v_k(t)$ before τ^k. Applying Proposition 4.3 with

$$\epsilon := \frac{a}{4C_1(K+2L)} \left(1 \wedge \frac{1}{4\mathscr{C}}\right), \tag{5.17}$$

where C_1 is the constant in Proposition 4.3, we find $T_0 \ge 2$ and $N \ge 1$ depending on $a, \nu, h, \mathcal{B}_1, \mathcal{B}_\varphi$ such that (4.1) holds. If $\tau^k \le T_0$, then by proceeding similarly as in the derivation of (4.21),

$$\|g_k(t)\|^2 \le C\|g_k(0)\|^2 \exp\left(-at + C_{a,\nu} \int_0^t \left(\|u_k\|_{H^1}^2 + \|v_k\|_{H^1}^2\right) \mathrm{d}s\right), \tag{5.18}$$

where $C > 0$ is a constant depending on $a, \nu, h, \mathcal{B}_1, \mathcal{B}_\varphi$. Notice that by the definition of τ^k,

$$\mathcal{E}_1^{u_k}(t) < (K+L)t + \rho + \epsilon' akT + \mathscr{C}\|u_k(0)\|^2, \tag{5.19}$$

$$\mathcal{E}_1^{v_k}(t) < (K+L)t + \rho + \epsilon' akT + \mathscr{C}\|v_k(0)\|^2 \tag{5.20}$$

for any $t < \tau^k$. Then, from (5.18)–(5.20), and the Young inequality, it follows

$$\begin{aligned} \|g_k(t)\|^2 &\le C\|g_k(0)\|^2 \\ &\quad \times \exp\left(-at + C[(K+L)T_0 + \rho + \epsilon' akT + \|u_k(0)\|^2 + \|v_k(0)\|^2]\right) \\ &\le C\|g_k(0)\|^2 \exp\left(-at + \frac{a(\|\tilde{u}(kT)\|^6 + \|\tilde{v}(kT)\|^6)}{16(K+2L)} + C\epsilon' akT\right), \end{aligned} \tag{5.21}$$

where C depends on $q, a, \nu, h, \mathcal{B}_1, \mathcal{B}_\varphi$. If $\tau^k > T_0$, it is clear that (5.21) holds up to T_0. Therefore, by Proposition 4.3 with ϵ given by (5.17), the definition of τ^k, and (5.21), we have

$$\begin{aligned} \|g_k(t)\|^2 &\le C\|g_k(T_0)\|^2 \exp\left(-\frac{at}{2} + \frac{a(\|u_k(0)\|^6 + \|v_k(0)\|^6)}{16(K+2L)} + C\epsilon' akT\right) \\ &\le C\|g_k(0)\|^2 \exp\left(-\frac{at}{2} + \frac{a(\|\tilde{u}(kT)\|^6 + \|\tilde{v}(kT)\|^6)}{8(K+2L)} + C\epsilon' akT\right). \end{aligned} \tag{5.22}$$

On the other hand, due to the definitions of $\mathcal{E}_\psi^{\tilde{u}}$, $\mathcal{E}_\psi^{\tilde{v}}$, and

$$\|\tilde{u}(kT)\|^6 \le \mathcal{E}_\psi^{\tilde{u}}(kT) \le (K+2L)kT + 2\rho + \mathscr{C}(1+d^6), \tag{5.23}$$

$$\|\tilde{v}(kT)\|^6 \le \mathcal{E}_\psi^{\tilde{v}}(kT) \le (K+2L)kT + 2\rho + \mathscr{C}(1+d^6), \tag{5.24}$$

and

$$\|g_k(0)\|^2 = \|\tilde{u}(kT) - \tilde{v}(kT)\|^2 \le Cd^2 \exp\left(-\frac{a}{2}kT + C(\rho + d^6)\right) \tag{5.25}$$

on $\{\sigma \ge kT\}$. Hence, by combining the estimates (5.21)–(5.25), we infer

$$\|g_k(t)\|^2 \le Cd^2 \exp\left(-\left(\frac{1}{4} - C\epsilon'\right) akT - \frac{a}{2}t + Cd^6\right) \tag{5.26}$$

for any $t \in [0, \tau^k)$ with the constant $C > 0$ depending on $q, a, \nu, h, \mathcal{B}_1, \mathcal{B}_\varphi$. Choosing $\epsilon' = \frac{1}{8C} \wedge 1$, we get

$$\|g_k(t)\|^2 \le Cd^2 \exp\left(-\frac{a}{8}kT - \frac{a}{2}t + Cd^6\right) \tag{5.27}$$

Moreover, as in (4.29),

$$\|\mathcal{A}_k(t)\|^2 \le C\mathbb{I}_{\{t \le \tau^k\}} \|g_k\|^2 \left(1 + \|u_k\|_{H^1}^2 + \|v_k\|_{H^1}^2\right), \tag{5.28}$$

where by (5.19), (5.21), (5.23), (5.24),

$$\begin{aligned}
\int_0^t (\|u_k\|_{H^1}^2 + \|v_k\|_{H^1}^2) \mathrm{d}s &\le \mathcal{E}_1^{u_k}(t) + \mathcal{E}_1^{v_k}(t) \\
&\le 2(K+L)t + 2\rho + 2\epsilon' akT + \mathscr{C}(\|\tilde{u}(kT)\|^2 + \|\tilde{v}(kT)\|^2) \\
&\le 2(K+L)t + 2\rho + 2akT + \mathscr{C}(\mathcal{E}_\psi^{\tilde{u}}(kT) + \mathcal{E}_\psi^{\tilde{v}}(kT)) \\
&\le 2(K+L)t + 2\rho + 2akT + \mathscr{C}[2(K+2L)kT + 4\rho + 2\mathscr{C}(1+d^6)] \\
&\le C(t + kT + d^6) + C \\
&\le C(1 + kT + d^6)(1+t)
\end{aligned} \tag{5.29}$$

for $t < \tau^k$. Proceeding similarly as in the derivation of the estimates (4.30) and (4.31) combined with an application of (5.28) and (5.29),

$$\int_0^\infty \|\mathcal{A}_k(t)\|^2 \mathrm{d}t \le Cd^2 e^{Cd^6} e^{-\frac{akT}{16}} \tag{5.30}$$

on the event $\{\sigma \ge kT\}$. Hence,

$$I^1 \le \left(\exp\left(Cd^2 e^{Cd^6} e^{-\frac{akT}{16}}\right) - 1\right)^{\frac{1}{2}}.$$

We choose $d < 1$ so small that

$$C_d := Cd^2 e^{Cd^6} \le 1.$$

Note that there is a constant $c > 0$ such that $e^x - 1 \le cx$ for any $x \in (0,1)$. Then, by choosing $T \ge \frac{32}{a}$, we derive

$$I^1 \lesssim \sqrt{C_d} e^{-k} \lesssim_q \sqrt{C_d}(k+1)^{-4q},$$

which implies

$$I^1 \leq \frac{q-1}{16q}(k+1)^{-4q}, \tag{5.31}$$

provided that d is sufficiently small. Next, we bound the terms I^2, I^3 in (5.16). For I^2, we apply Corollary 3.8 and (A.1) to derive

$$\begin{aligned} I^2 &= \mathbb{E}_{\boldsymbol{u}}\left(\mathbb{I}_{\{\sigma\geq kT\}}\mathbb{P}_{\boldsymbol{u}(kT)}\{\tau^u<\infty\}\right) \leq \mathbb{E}_{\boldsymbol{u}}\left(\mathbb{P}_{\boldsymbol{u}(kT)}\{\tau^u<\infty\}\right) \\ &\leq C\left(1+\mathbb{E}_{\boldsymbol{u}}\|\tilde{u}(kT)\|^{8(8q+3)}\right)\left(e^{-\gamma(\rho+\epsilon' akT)}+\frac{1}{(\rho+\epsilon' akT)^{4q}}\right) \\ &\lesssim_{q,a,\nu,h,\mathcal{B}_1,\mathcal{B}_\varphi} \frac{1}{(\rho+\epsilon' akT)^{4q}}. \end{aligned} \tag{5.32}$$

By similarly choosing the constants ρ, T as in the derivation of (5.11), we obtain

$$I^2 \leq \frac{q-1}{16q}(k+1)^{-4q}. \tag{5.33}$$

The term I^3 is estimated similarly, with the help of Corollary 3.8. This completes the proof. □

We turn to the verification of properties (2.4)–(2.7). Let $N \geq 1$ be such that Lemma 5.5, along with Propositions 5.4 and 4.3, holds for ϵ as defined in (5.17). Let $q > 1$ be arbitrary, and let d, T, ρ, L be the constants in Lemma 5.5. The property (2.4) follows from the definition of σ. Applying Lemma 5.5, we obtain

$$\mathbb{P}_{\boldsymbol{u}}\{\sigma=\infty\} \geq 1-\sum_{k=0}^{\infty}\mathbb{P}_{\boldsymbol{u}}\{\sigma\in[kT,(k+1)T]\} \geq \frac{1}{2} > 0,$$

which establishes (2.5). To prove (2.6), we derive from Lemma 5.5 that

$$\begin{aligned} \mathbb{E}_{\boldsymbol{u}}\left(\mathbb{I}_{\{\sigma<\infty\}}\sigma^q\right) &\leq \sum_{k=0}^{\infty}\mathbb{E}_{\boldsymbol{u}}\left(\mathbb{I}_{\{\sigma\in[kT,(k+1)T]\}}\sigma^q\right) \\ &\lesssim_{q,T} \sum_{k=0}^{\infty}(k+1)^q\mathbb{P}_{\boldsymbol{u}}\{\sigma\in[kT,(k+1)T]\} \\ &\lesssim_{q,T} \sum_{k=0}^{\infty}(k+1)^{-3q} \leq C_q. \end{aligned}$$

Lastly, to check (2.7), using the definition of σ, we note that

$$\mathbb{E}_{\boldsymbol{u}}\left(\mathbb{I}_{\{\sigma<\infty\}}\left(\|\tilde{u}(\sigma)\|^{2q}+\|\tilde{u}'(\sigma)\|^{2q}\right)\right) \leq C_q\mathbb{E}_{\boldsymbol{u}}(\mathbb{I}_{\{\sigma<\infty\}}(1+\sigma^q)) \leq C_q.$$

This completes the proof of polynomial squeezing, thereby completing the proof of Theorem 2.2.

A Moment estimates

The following lemma gathers standard a priori estimates for the stochastic NS system (1.1); see Chapter 2 in [KS12] for more general results.

Lemma A.1. *Let $u(t)$ be a solution of (1.1), and let $w(t)$ be the corresponding vorticity. Then, for any $p \geq 1$, any integer $m \geq 1$, and time $T > 0$, we have*

$$\mathbb{E}\|u(T)\|^{2p} \leq e^{-apT}\mathbb{E}\|u_0\|^{2p} + C_{a,p,h,\mathcal{B}_0}, \tag{A.1}$$

$$\mathbb{E}(\mathcal{E}_p^u(T))^m \leq C_{a,\nu,h,\mathcal{B}_0,p,m}\left(\mathbb{E}\|u_0\|^{4p(m+2)} + T^m + 1\right), \tag{A.2}$$

$$\mathbb{E}\|w(T)\|^{2p} \leq C_{a,\nu,h,\mathcal{B}_1,p,T}\left(\mathbb{E}\|u_0\|^{4(p+2)} + 1\right), \tag{A.3}$$

where $\mathcal{E}_p^u$ is defined in (3.3).

Proof. First, let us establish the estimate (A.1). Taking the expectation in (3.4) and using the Young inequality, we get

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}\|u(t)\|^{2p} + 2p\nu\mathbb{E}\|\nabla u(t)\|^2\|u(t)\|^{2p-2} + ap\mathbb{E}\|u(t)\|^{2p} \leq C_{a,p,h,\mathcal{B}_0}.$$

Applying the Gronwall inequality, we arrive at the required estimate.

Next, we turn to the estimate (A.2). Note that if X_1 and X_2 are non-negative random variables, then

$$\begin{aligned}\mathbb{E}X_1^m &\leq 2^m\mathbb{E}((X_1 - X_2)^m\mathbb{I}_{\{X_1 \geq X_2\}}) + 2^m\mathbb{E}X_2^m \\ &\leq 2^m\int_0^\infty \mathbb{P}\{X_1 - X_2 \geq \rho^{\frac{1}{m}}\}\mathrm{d}\rho + 2^m\mathbb{E}X_2^m.\end{aligned} \tag{A.4}$$

Applying this inequality with

$$X_1 := \mathcal{E}_p^u(T), \qquad X_2 := \kappa_p T + \mathcal{C}_p\|u_0\|^{2p},$$

where $\kappa_p, \mathcal{C}_p$ are given in Proposition 3.1, we get

$$\begin{aligned}\mathbb{E}(\mathcal{E}_p^u(T))^m \leq 2^m\int_0^\infty \mathbb{P}\left\{\sup_{t\geq 0}\left(\mathcal{E}_p^u(t) - \kappa_p t - \mathcal{C}_p\|u_0\|^{2p}\right) \geq \rho^{\frac{1}{m}}\right\}\mathrm{d}\rho \\ + 4^m\left(\mathcal{C}_p^m\mathbb{E}\|u_0\|^{2pm} + \kappa_p^m T^m\right).\end{aligned}$$

By choosing $q := 2(m+2)$ in Proposition 3.1, we obtain

$$\begin{aligned}\mathbb{E}(\mathcal{E}_p^u(T))^m &\leq 2^m C_{a,\nu,h,\mathcal{B}_0,p,m}\left(\mathbb{E}\|u_0\|^{(2p-1)(2m+4)} + 1\right)\int_2^\infty \frac{1}{\rho^{1+\frac{1}{m}}}\mathrm{d}\rho \\ &\quad + 2^{m+1} + 4^m\left(\mathcal{C}_p^m\mathbb{E}\|u_0\|^{2pm} + \kappa_p^m T^m\right) \\ &\leq C_{a,\nu,h,\mathcal{B}_0,p,m}\left(\mathbb{E}\|u_0\|^{4p(m+2)} + T^m + 1\right)\end{aligned}$$

as desired. Finally, to prove (A.3), we apply the Itô formula and use (3.9):

$$\begin{aligned}\mathrm{d}(t\|w(t)\|^2) = 2t\langle w, \nu\Delta w - aw + \operatorname{curl} h\rangle \mathrm{d}t + \mathrm{d}\tilde{M}_1(t)\\ + t\sum_{j=1}^{\infty} b_j^2 \|\operatorname{curl} e_j\|^2 \mathrm{d}t + \|w\|^2 \mathrm{d}t,\end{aligned}$$

where

$$\tilde{M}_1(t) := 2\int_0^t s\sum_{j=1}^{\infty} b_j \langle w, \operatorname{curl} e_j\rangle \mathrm{d}\beta_j(s).$$

Notice that for any $t \le T$,

$$\langle \tilde{M}_1\rangle(t) := 4\int_0^t s^2 \sum_{j=1}^{\infty} b_j^2 \langle w, \operatorname{curl} e_j\rangle^2 \mathrm{d}s \le 4T\mathcal{B}_1 \int_0^t s\|w(s)\|^2 \mathrm{d}s.$$

By setting $\gamma^* := \frac{a}{8T\mathcal{B}_1}$ and using the Young inequality, we obtain

$$\begin{aligned}t\|w(t)\|^2 + \int_0^t s\left(a\|w(s)\|^2 + 2\nu\|\nabla w(s)\|^2\right)\mathrm{d}s\\ \le C_{a,h,\mathcal{B}_1} t^2 + \int_0^t \|w(s)\|^2 \mathrm{d}s + \tilde{M}_1(t) - \frac{\gamma^*}{2}\langle \tilde{M}_1\rangle(t),\end{aligned}$$

which implies

$$t\|w(t)\|^2 \le C_{a,h,\mathcal{B}_1} T^2 + C\mathcal{E}_1^u(t) + \tilde{M}_1(t) - \frac{\gamma^*}{2}\langle \tilde{M}_1\rangle(t)$$

for any $0 < t \le T$. Applying the inequality (A.4) with

$$X_1 := t\|w(t)\|^2, \qquad X_2 := C_{a,h,\mathcal{B}_1} T^2 + C\mathcal{E}_1^u(t),$$

the exponential supermartingale estimate, and (A.2), we derive

$$\begin{aligned}\mathbb{E}(t^p\|w(t)\|^{2p}) &\le 2^p \int_0^{\infty} \mathbb{P}\left\{\sup_{t\ge 0}\left(\tilde{M}_1(t) - \frac{\gamma^*}{2}\langle \tilde{M}_1\rangle(t)\right) \ge \rho^{\frac{1}{p}}\right\}\mathrm{d}\rho\\ &\quad + C_{a,\nu,h,\mathcal{B}_1,p,T}\left(1 + \mathbb{E}\left(\mathcal{E}_1^u(t)\right)^p\right)\\ &\le C_{a,\nu,h,\mathcal{B}_1,p,T}\left(1 + \mathbb{E}\|u_0\|^{4(p+2)}\right).\end{aligned}$$

Setting $t = T$ in this inequality, we get the desired estimate (A.3). □

B Weighted estimate for the Leray projector

In this subsection, we prove Lemma 4.2. The proof of the property (a) relies on weighted estimates for singular integral operators. Let us begin by recalling the following definition.

Definition B.1. A locally integrable function $w : \mathbb{R}^n \to \mathbb{R}_+$ is said to be a Muckenhoupt A_2 weight, if

$$[w]_{A_2} := \sup_{\text{all balls } B} \left(\frac{1}{|B|}\int_B w\mathrm{d}x\right)\left(\frac{1}{|B|}\int_B w^{-1}\mathrm{d}x\right) < \infty,$$

where $|B|$ denotes the Lebesgue measure of B.

We will use the following weighted estimates for the Riesz transform established in [Pet08].

Theorem B.2. *There exists a constant $C > 0$ such that for any $w \in A_2$, the Riesz transforms $\mathcal{R}_j$, $j = 1, 2, \ldots, n$ defined by (3.18) as operators in weighted space $\mathcal{R}_j : L^2(\mathbb{R}^n, w\mathrm{d}x) \to L^2(\mathbb{R}^n, w\mathrm{d}x)$ have operator norm satisfying the estimate*

$$\|\mathcal{R}_j\| \le C[w]_{A_2}.$$

Before proving the property (a), let us establish a uniform-in-time bound for the quantity $[\psi(t)]_{A_2}$.

Lemma B.3. *Let ψ be the space-time weight function given by (3.2). Then*

$$\sup_{t\ge 2}[\psi(t)]_{A_2} < \infty.$$

Proof. Step 1. In this step, we derive pointwise upper and lower bounds for ψ. Define

$$f_1(y) := 1 - e^{-y}, \qquad f_2(y) := \frac{1-e^{-y}}{y}, \qquad y > 0.$$

The function f_1 is strictly increasing, and f_2 is strictly decreasing on $(0, \infty)$. Therefore, in the ball $\{x \,|\, |x| \le \sqrt{t^2-1}\}$, we have

$$t \ge \varphi \ge \psi(t,x) = \varphi(1 - e^{-t/\varphi}) \ge \varphi(1 - e^{-1}).$$

Similarly,

$$\varphi \ge t \ge \psi(t,x) = t\frac{\varphi}{t}(1 - e^{-t/\varphi}) \ge t(1 - e^{-1})$$

in the outer region $\{x \,|\, |x| > \sqrt{t^2-1}\}$. To summarize, we have

$$(1 - e^{-1})(t \wedge \varphi) \le \psi(t,x) \le t \wedge \varphi, \qquad \forall t \ge 2, x \in \mathbb{R}^2.$$

Thus, it suffices to establish

$$\sup_{t\ge 2}[t \wedge \varphi]_{A_2} < \infty, \tag{B.1}$$

due to the definition of Muckenhoupt A_2 weights.
Step 2. We claim that

$$\sup_{t\ge 2}\sup_{R>0}\left(\frac{1}{|B(0,R)|}\int_{B(0,R)} t \wedge \varphi\mathrm{d}x\right)\left(\frac{1}{|B(0,R)|}\int_{B(0,R)} (t \wedge \varphi)^{-1}\mathrm{d}x\right) < \infty. \tag{B.2}$$

For $R \geq \sqrt{t^2-1}$, we compute the integral by using polar coordinates:

$$\begin{aligned}\int_{B(0,R)} (t \wedge \varphi)^{-1} \mathrm{d}x &= 2\pi \int_0^R (t \wedge \sqrt{r^2+1})^{-1} r \mathrm{d}r \\ &= 2\pi \int_0^{\sqrt{t^2-1}} \frac{r}{\sqrt{r^2+1}} \mathrm{d}r + 2\pi \int_{\sqrt{t^2-1}}^R t^{-1} r \mathrm{d}r \\ &= 2\pi \left((r^2+1)^{\frac{1}{2}} \big|_0^{\sqrt{t^2-1}} + \frac{r^2}{2t} \big|_{\sqrt{t^2-1}}^R \right) \\ &= 2\pi \left(\frac{t}{2} - 1 + \frac{1}{2t} + \frac{R^2}{2t} \right) = \frac{\pi}{t}(R^2 + (t-1)^2). \end{aligned} \tag{B.3}$$

Similarly,

$$\begin{aligned}\int_{B(0,R)} t \wedge \varphi \mathrm{d}x &= 2\pi \int_0^R (t \wedge \sqrt{r^2+1}) r \mathrm{d}r \\ &= 2\pi \int_0^{\sqrt{t^2-1}} r\sqrt{r^2+1} \mathrm{d}r + 2\pi \int_{\sqrt{t^2-1}}^R t r \mathrm{d}r \\ &= 2\pi \left(\frac{1}{3}(r^2+1)^{\frac{3}{2}} \big|_0^{\sqrt{t^2-1}} + \frac{tr^2}{2} \big|_{\sqrt{t^2-1}}^R \right) \\ &= 2\pi \left(-\frac{1}{6}t^3 + \frac{t}{2} - \frac{1}{3} + \frac{tR^2}{2} \right) \\ &\leq \pi t R^2, \end{aligned} \tag{B.4}$$

where we used

$$-\frac{1}{6}t^3 + \frac{t}{2} - \frac{1}{3} \leq -\frac{2}{3} < 0, \qquad \forall t \geq 2.$$

Plugging the estimates (B.3) and (B.4) into (B.2), we obtain

$$\begin{aligned}&\left(\frac{1}{|B(0,R)|} \int_{B(0,R)} t \wedge \varphi \mathrm{d}x \right) \left(\frac{1}{|B(0,R)|} \int_{B(0,R)} (t \wedge \varphi)^{-1} \mathrm{d}x \right) \\ &\qquad \leq 1 + \frac{(t-1)^2}{R^2} \leq 1 + \frac{(t-1)^2}{t^2-1} \leq 2 \end{aligned} \tag{B.5}$$

for any $t \geq 2$ and $R \geq \sqrt{t^2-1}$. As for the case $R < \sqrt{t^2-1}$, we similarly have

$$\int_{B(0,R)} (t \wedge \varphi)^{-1} \mathrm{d}x = 2\pi(\sqrt{R^2+1} - 1)$$

and

$$\int_{B(0,R)} (t \wedge \varphi) \mathrm{d}x = \frac{2\pi}{3}((R^2+1)^{\frac{3}{2}} - 1),$$

which implies

$$\left(\frac{1}{|B(0,R)|}\int_{B(0,R)} t\wedge\varphi \mathrm{d}x\right)\left(\frac{1}{|B(0,R)|}\int_{B(0,R)} (t\wedge\varphi)^{-1}\mathrm{d}x\right) = \frac{4}{3}\frac{(\sqrt{R^2+1}-1)((R^2+1)^{\frac{3}{2}}-1)}{R^4} =: G(R).$$

Noticing that

$$\lim_{R\to\infty} G(R) = \frac{4}{3}, \qquad \lim_{R\to 0^+} G(R) = 1,$$

we conclude that

$$\left(\frac{1}{|B(0,R)|}\int_{B(0,R)} t\wedge\varphi \mathrm{d}x\right)\left(\frac{1}{|B(0,R)|}\int_{B(0,R)} (t\wedge\varphi)^{-1}\mathrm{d}x\right) \le \sup_{R>0} G(R) < \infty \tag{B.6}$$

for any $t \ge 2$ and $R < \sqrt{t^2-1}$. Combining (B.5) and (B.6), we obtain the claimed estimate (B.2).

Step 3. To obtain the estimate (B.1), let us divide all balls $B(x_0,R) \subset \mathbb{R}^2$ into two types: type I if $|x_0| \ge 3R$ and type II if $|x_0| < 3R$. In this step, we derive bounds for type I balls. For any $x \in B(x_0,R)$, there holds

$$\frac{2|x_0|}{3} \le |x_0| - R \le |x| \le |x_0| + R \le \frac{4|x_0|}{3},$$

which implies

$$\left(\frac{1}{|B(x_0,R)|}\int_{B(x_0,R)} t\wedge\varphi \mathrm{d}x\right)\left(\frac{1}{|B(x_0,R)|}\int_{B(x_0,R)} (t\wedge\varphi)^{-1}\mathrm{d}x\right) \le \frac{t\wedge\sqrt{1+(4|x_0|/3)^2}}{t\wedge\sqrt{1+(2|x_0|/3)^2}} =: F(t,x_0).$$

Furthermore,

$$F(t,x_0) = \begin{cases} \frac{\sqrt{1+(4|x_0|/3)^2}}{\sqrt{1+(2|x_0|/3)^2}}, & \text{if } \sqrt{1+(4|x_0|/3)^2} \le t, \\ \frac{t}{\sqrt{1+(2|x_0|/3)^2}}, & \text{if } \sqrt{1+(2|x_0|/3)^2} \le t < \sqrt{1+(4|x_0|/3)^2}, \\ 1, & \text{if } t < \sqrt{1+(2|x_0|/3)^2}. \end{cases} \tag{B.7}$$

It is clear that

$$\sup_{y>0} \frac{\sqrt{1+(4y/3)^2}}{\sqrt{1+(2y/3)^2}} < \infty. \tag{B.8}$$

Notice that if $\sqrt{1+(2|x_0|/3)^2} \le t < \sqrt{1+(4|x_0|/3)^2}$, then

$$\frac{t^2-1}{4} \le \frac{4|x_0|^2}{9},$$

which ensures the estimate

$$\frac{t}{\sqrt{1+(2|x_0|/3)^2}} \le \frac{t}{\sqrt{1+\frac{t^2-1}{4}}} \le \sup_{t\ge 2} \frac{t}{\sqrt{1+\frac{t^2-1}{4}}}. \tag{B.9}$$

Combining the estimates (B.7)–(B.9), we conclude

$$\sup_{t\ge 2} \sup_{\text{Type I balls}} \left(\frac{1}{|B|}\int_B t\wedge\varphi \mathrm{d}x\right)\left(\frac{1}{|B|}\int_B (t\wedge\varphi)^{-1}\mathrm{d}x\right) < \infty. \tag{B.10}$$

Step 4. As for type II balls, we note that

$$B(x_0,R) \subset B(0,4R).$$

Therefore, we arrive at

$$\begin{aligned}
&\left(\frac{1}{|B(x_0,R)|}\int_{B(x_0,R)} t\wedge\varphi \mathrm{d}x\right)\left(\frac{1}{|B(x_0,R)|}\int_{B(x_0,R)} (t\wedge\varphi)^{-1}\mathrm{d}x\right)\\
&\quad\le \left(\frac{16}{|B(0,4R)|}\int_{B(0,4R)} t\wedge\varphi \mathrm{d}x\right)\left(\frac{16}{|B(0,4R)|}\int_{B(0,4R)} (t\wedge\varphi)^{-1}\mathrm{d}x\right)\\
&\quad\lesssim \sup_{t\ge 2}\sup_{R>0}\left(\frac{1}{|B(0,R)|}\int_{B(0,R)} t\wedge\varphi \mathrm{d}x\right)\left(\frac{1}{|B(0,R)|}\int_{B(0,R)} (t\wedge\varphi)^{-1}\mathrm{d}x\right),
\end{aligned}$$

which implies

$$\sup_{t\ge 2} \sup_{\text{Type II balls}} \left(\frac{1}{|B|}\int_B t\wedge\varphi \mathrm{d}x\right)\left(\frac{1}{|B|}\int_B (t\wedge\varphi)^{-1}\mathrm{d}x\right) < \infty,$$

due to the estimate (B.2). This, together with the estimate (B.10), completes the proof. □

Theorem B.2 and Lemma B.3 imply that there exists a constant $C>0$ such that

$$\|\psi(t)\mathcal{R}_j f\| \le C\|\psi(t) f\|, \qquad \forall t \ge 2. \tag{B.11}$$

Moreover, for any vector field $f \in L^2(\mathbb{R}^2;\mathbb{R}^2)$, the Leray projector Π can be represented as follows:

$$(\Pi f)_i = f_i + \sum_{j=1}^{2} \mathcal{R}_i\mathcal{R}_j f_j.$$

Combining this equality with the estimate (B.11), we obtain the property (a).

To prove (b), note that the assumption $\operatorname{div} u = 0$ implies that

$$-\Delta\partial_1 u = \partial_1 \nabla^\perp w.$$

Taking the weighted L^2-inner product of this equation with $\partial_1 u$, we get

$$-\langle \psi\partial_1 u, \psi\Delta\partial_1 u\rangle = \langle \psi\partial_1 u, \psi\partial_1\nabla^\perp w\rangle,$$

where

$$-\langle \psi\partial_1 u, \psi\Delta\partial_1 u\rangle = \|\psi\nabla\partial_1 u\|^2 + \sum_{j=1}^{2} 2\langle \nabla\psi\partial_1 u_j, \psi\nabla\partial_1 u_j\rangle$$

and

$$\langle \psi\partial_1 u, \psi\partial_1\nabla^\perp w\rangle = \langle \operatorname{curl}(\psi^2\partial_1 u), \partial_1 w\rangle = \|\psi\partial_1 w\|^2 + 2\langle \nabla\psi(\partial_1 u)^\perp, \psi\partial_1 w\rangle.$$

Here, $(\partial_1 u)^\perp := (\partial_1 u_2, -\partial_1 u_1)$. Therefore,

$$\|\psi\nabla\partial_1 u\| \lesssim \|\nabla u\| + \|\psi\nabla w\|.$$

The weighted estimate for $\nabla\partial_2 u$ is proved in the same way.

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